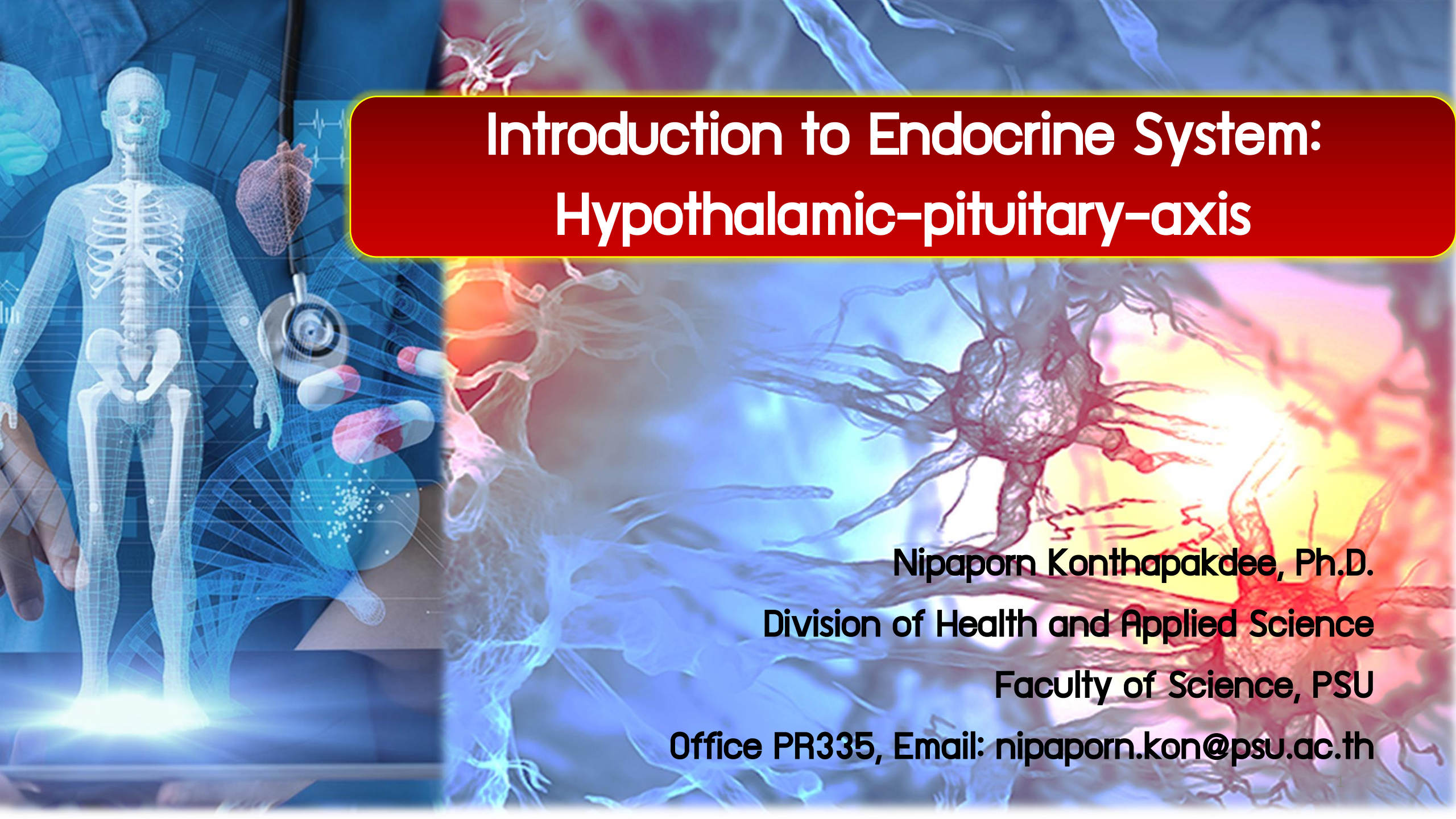


Introduction to Endocrine System: Hypothalamic-pituitary-axis

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Introduction Endocrine System: Hypothalamic-pituitary-axis (1 hr)

Learning objectives

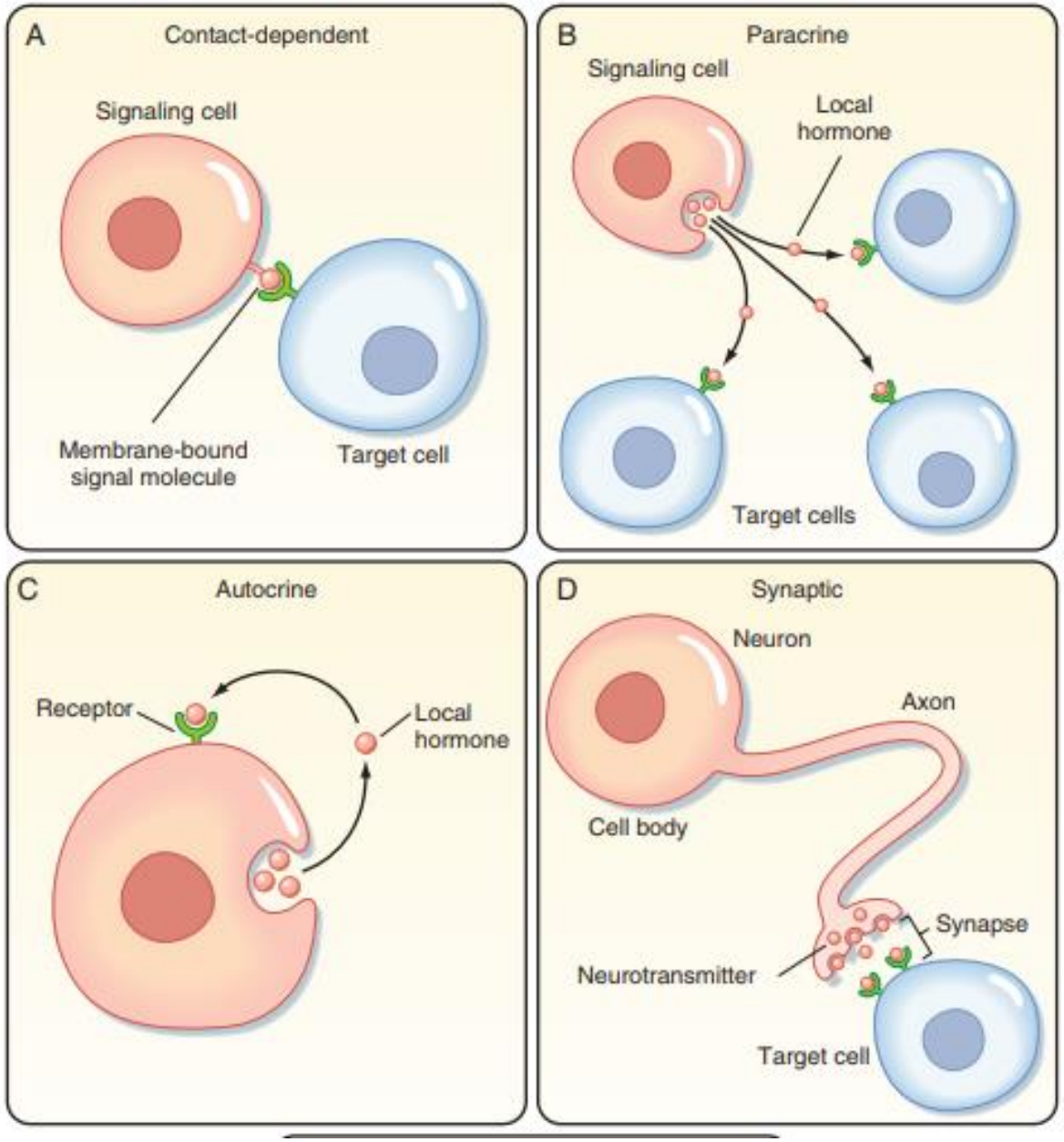
1. บอกชื่อและตำแหน่งของต่อมในระบบฮอร์โมนได้
2. บอกหน้าที่ทั่วไปของแต่ละต่อมได้
3. แบ่งประเภทตามโครงสร้างของฮอร์โมนแต่ละชนิดได้
4. อธิบายการสังเคราะห์ peptide and steroid hormones ได้
5. อธิบายรูปแบบการหลั่ง การขนส่งและการกำจัดออกจากร่างกาย
6. อธิบายกลไก feedback mechanism ในระบบต่อมไร้ท่อได้
7. อธิบายกลไกการออกฤทธิ์ของฮอร์โมนในระดับเซลล์ผ่าน second messenger
8. อธิบายหลักการของการควบคุมการหลั่งฮอร์โมนผ่าน Hypothalamic-pituitary-axis ได้

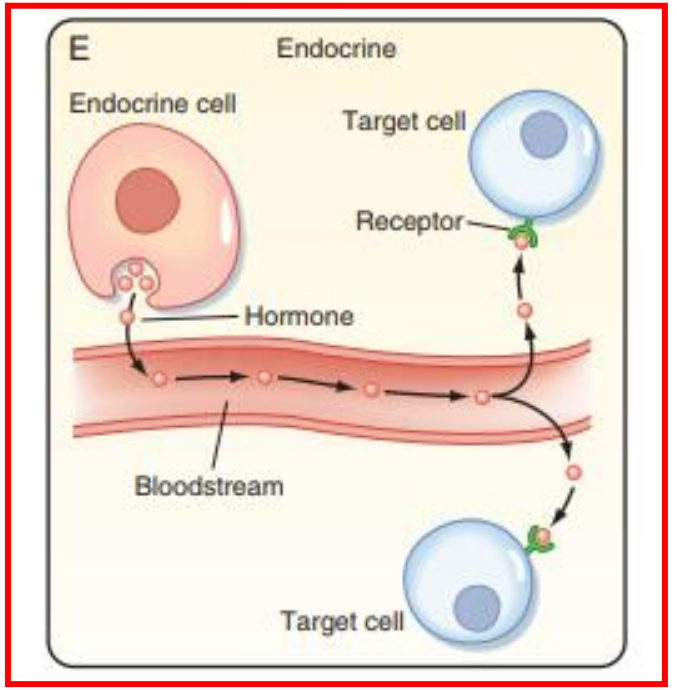
Introduction Endocrine System: Hypothalamic-pituitary-axis (1 hr)

Outlines

1. Basic components and functions of the endocrine system
2. Chemical structures of hormone
3. Synthesis of peptide and steroid hormones
4. Transport, metabolism and excretion of hormone
5. Types of feedback mechanisms
6. Intracellular signaling of hormone receptor activation
7. Second messengers
8. Hypothalamic-pituitary-axis

Cell-to-cell Communication (SDL)



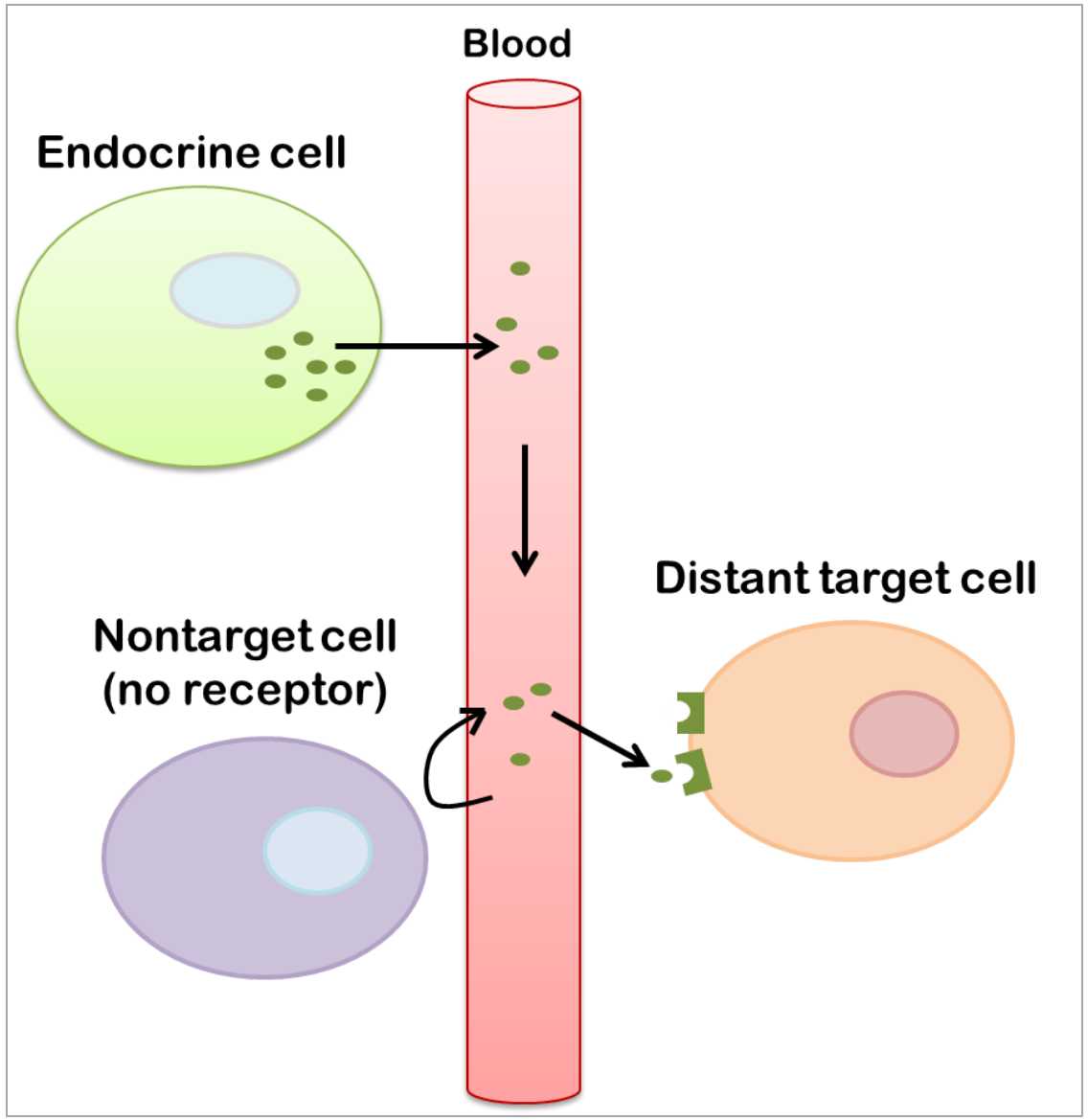


Autocrine : cell secretes a hormone or chemical messenger that binds to autocrine receptors on that same cell

Paracrine : cell produces a hormone or chemical messenger to induce changes in nearby cells

(จาก Koeppen and Stanton, 2018. Berne and Levy Physiology. หน้า 38)

Basic components of the endocrine system (SDL)



1. Endocrine gland
2. Hormone
3. Target organ

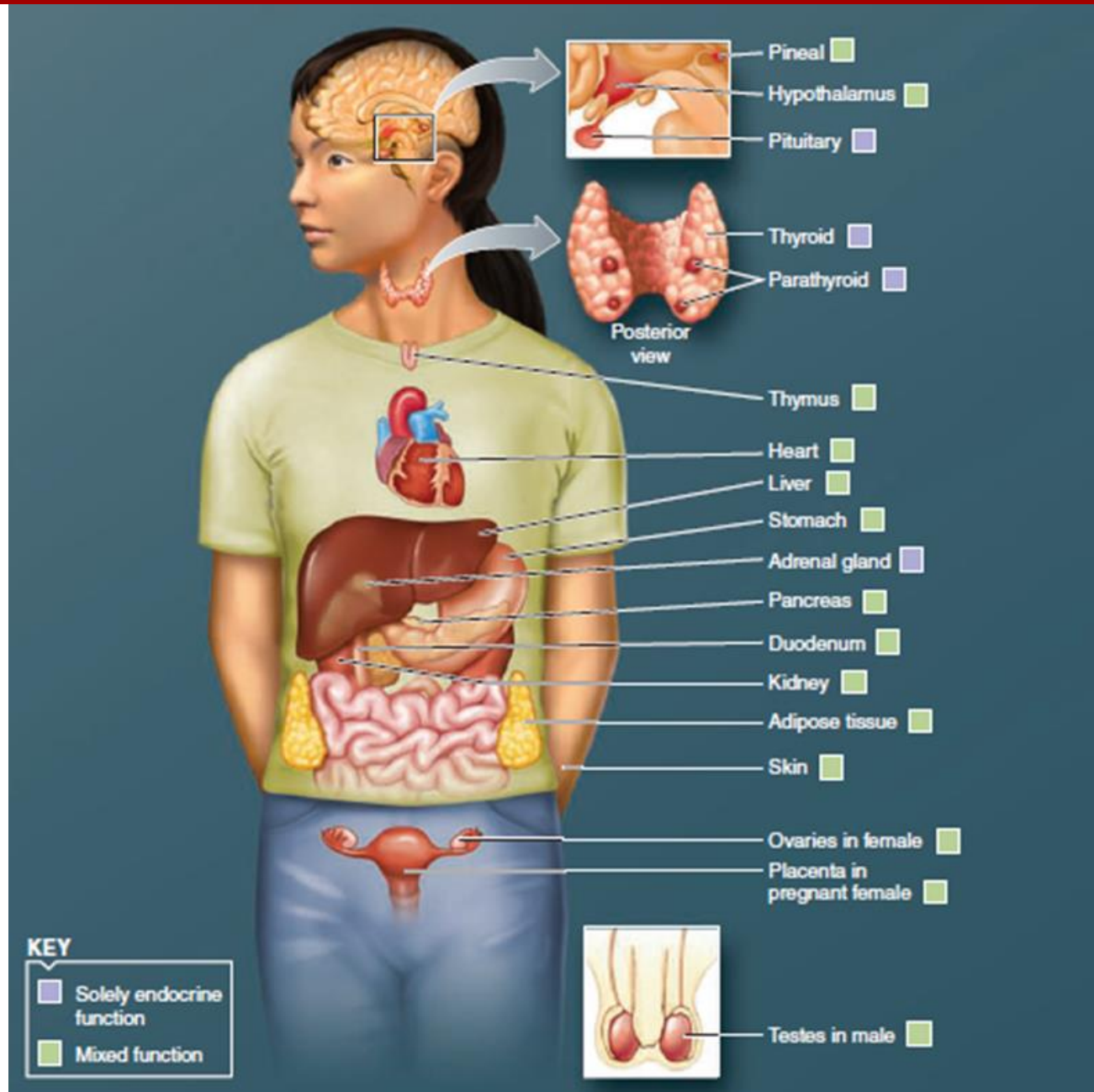
Endocrine glands in human body (SDL)

Hormones synthesized and secreted by dedicated endocrine glands

(i.e., Hypothalamus, Pituitary gland, Thyroid gland, Adrenal glands, Pancreases, Parathyroid glands)

Hormones Synthesized in organs with a primary function other than endocrine

(i.e., Heart, Thymus, Digestive tract, Gonads, Kidneys)



(จาก Sherwood, 2016. Human physiology. หน้า 639)

Hormone Actions (SDL)

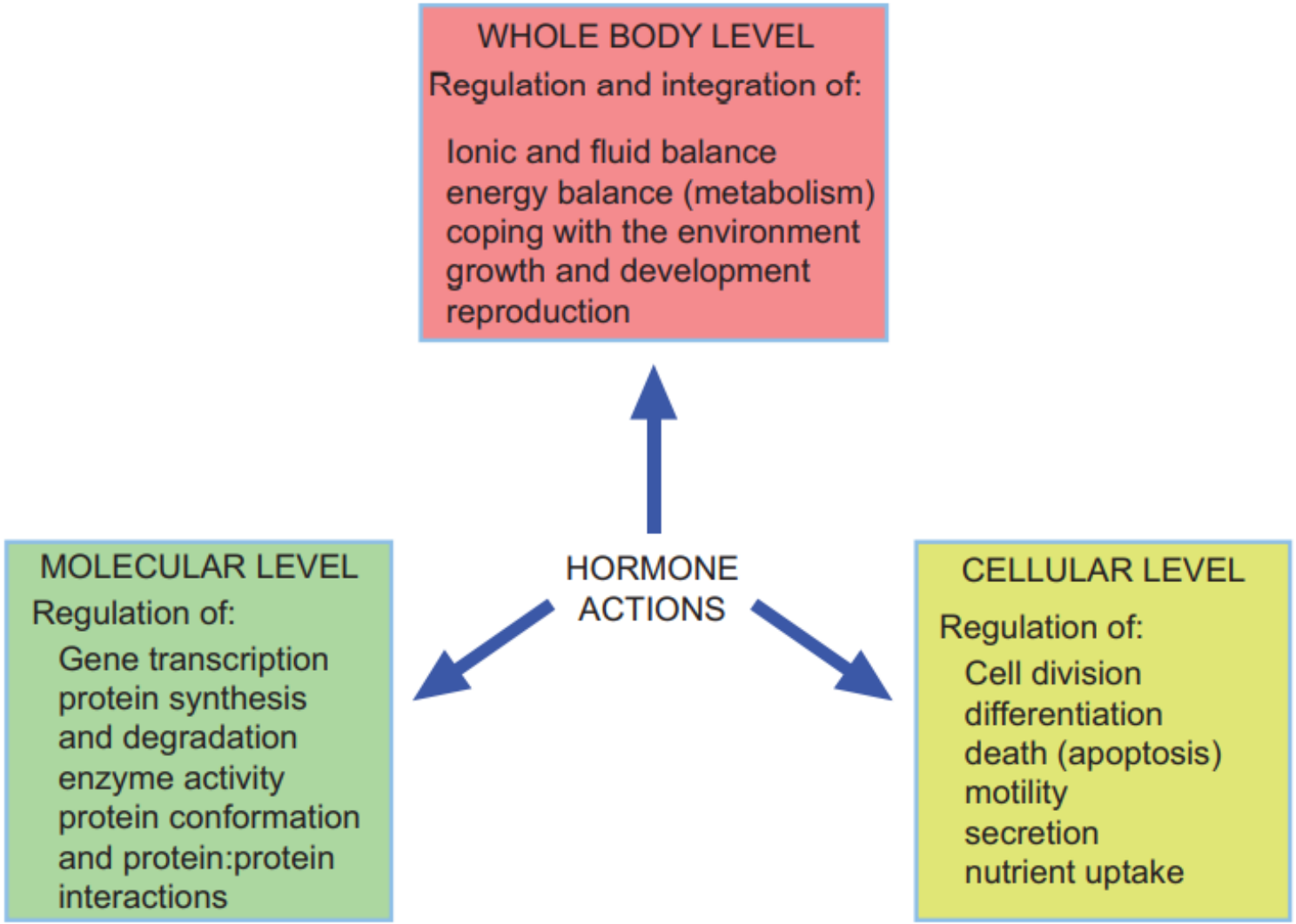


FIGURE 1.2 Levels at which hormone actions are considered.

(จาก Goodman, 2009. Basic medical endocrinology. หน้า 3)

Chemical Classification of Hormones (SDL)

1. Protein/Peptide hormone

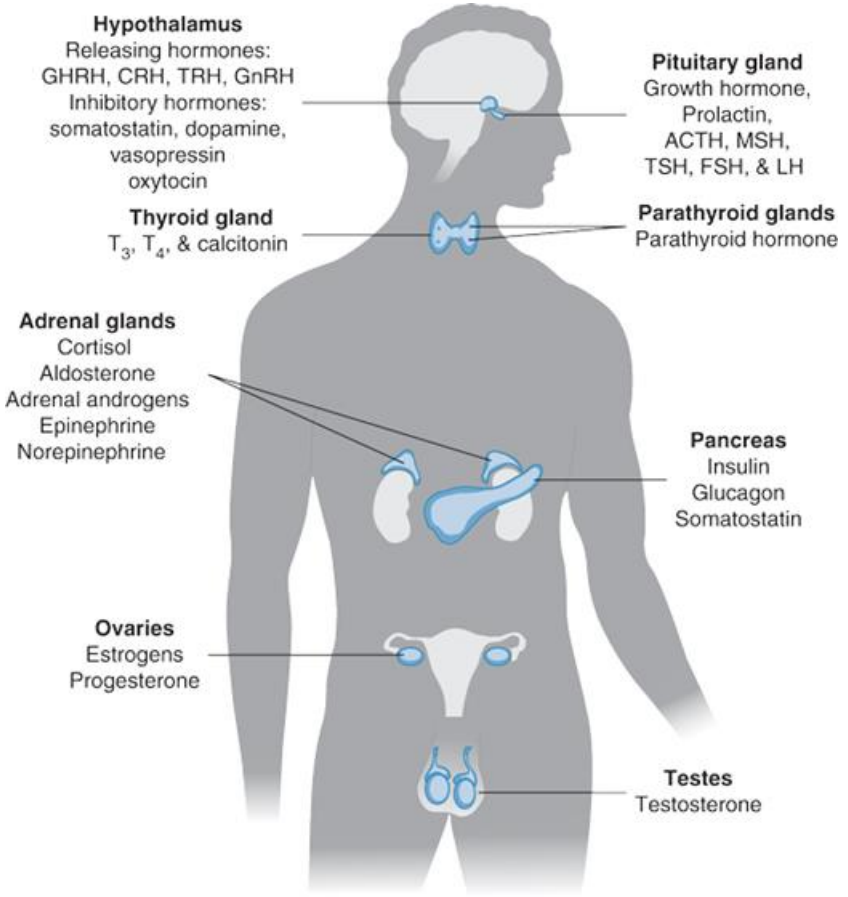
- molecules ranging from 3–200 amino acid residues
- **hormones from anterior and posterior pituitary gland, pancreas, and parathyroid gland**

2. Steroid hormone

- derived from cholesterol
- **hormones from adrenal cortex, gonad, and placenta**

3. Derivative of amino acid tyrosine

- synthesized from the amino acid tyrosine
- **hormones from thyroid gland and adrenal medulla**



Source: Molina PE: *Endocrine Physiology*, 4th Edition: www.accessmedicine.com
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Hormone structure influences the half-life of the hormone.

- Amines have the shortest half-life (2–3 minutes), polypeptides (4–40 minutes)
- steroids and proteins (4–170 minutes), thyroid hormones (0.75–6.7 days)

Chemical Classification of Hormones (SDL)

TABLE 1.2 Chemical Nature of the Classic Hormones

Tyrosine derivatives	Steroids	Peptides (<20 amino acids)	Proteins (>20 amino acids)
Epinephrine	Testosterone	Oxytocin	Insulin
Norepinephrine	Estradiol	Vasopressin	Glucagon
Dopamine	Progesterone	Angiotensin	Adrenocorticotropic hormone
Triiodothyronine	Cortisol	Melanocyte-stimulating hormone	Thyroid-stimulating hormone
Thyroxine	Aldosterone Vitamin D	Somatostatin Thyrotropin-releasing hormone Gastrin Cholecystokinin	Secretin Motilin Follicle-stimulating hormone Luteinizing hormone Gonadotropin-releasing hormone Growth hormone Prolactin Corticotropin-releasing hormone Growth hormone-releasing hormone Parathyroid hormone Calcitonin Chorionic gonadotropin Choriosomatomammotropin

(จาก Goodman, 2009. Basic medical endocrinology. หน้า 4)

Synthesis and Secretion of Peptide Hormones

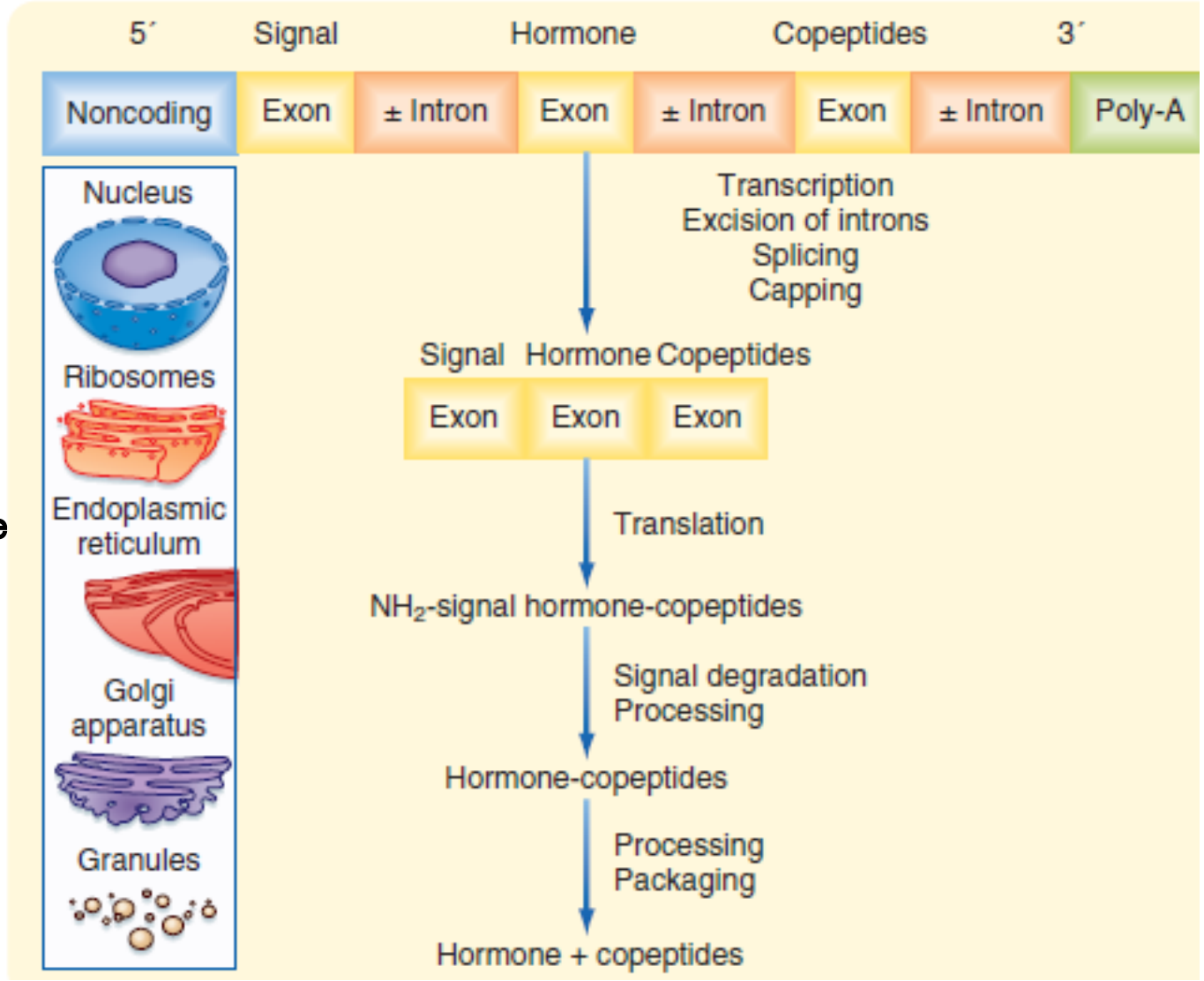
DNA

mRNA

Preprohormone

Prohormone

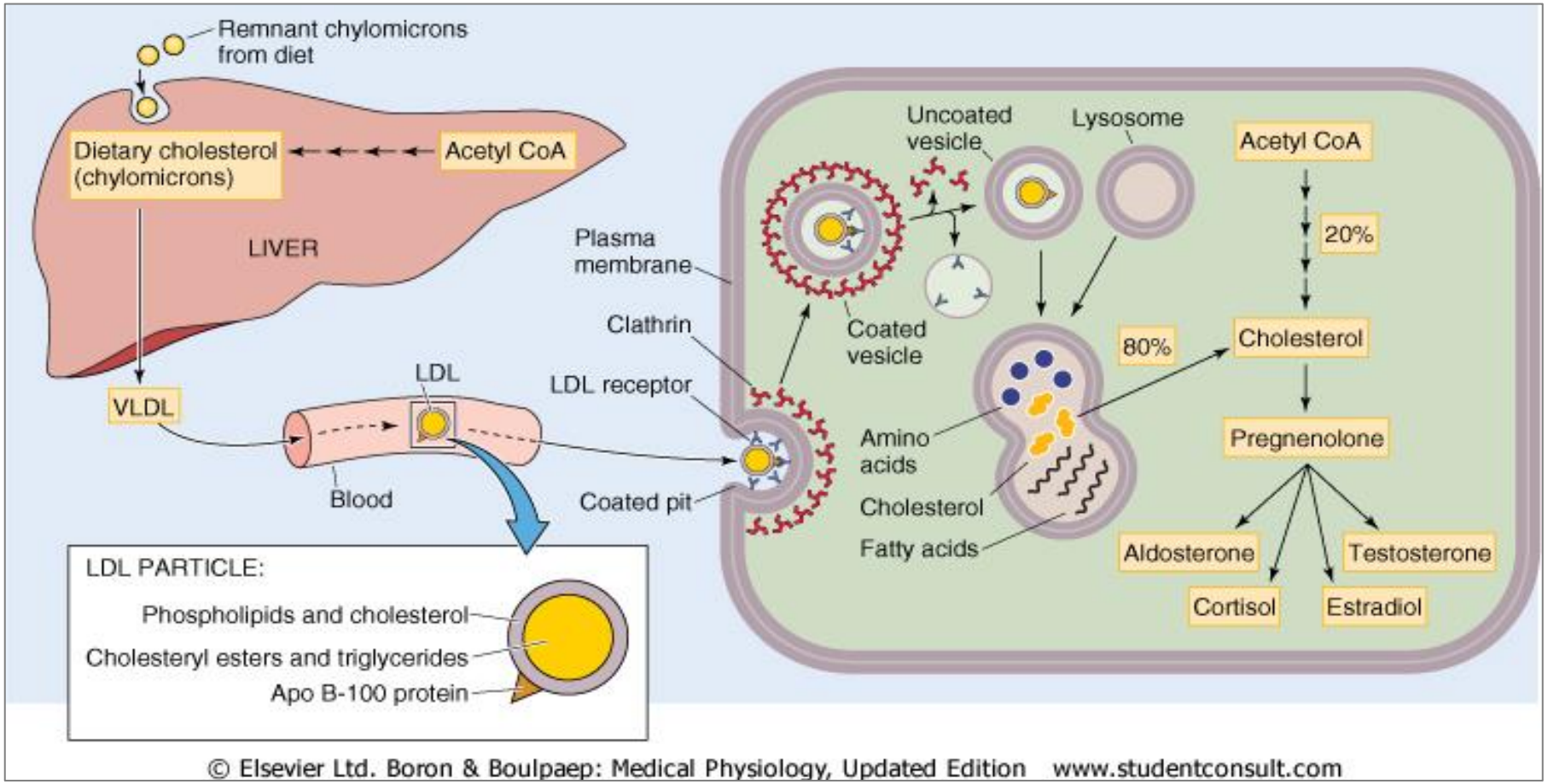
Hormone



- Peptide hormones are synthesized as **preprohormones** in the ribosomes and are processed to **prohormone** in the **endoplasmic reticulum (ER).**
- In the **Golgi apparatus**, the hormone or prohormone is packaged in **secretory vesicles**, which are released from the cell in response to an influx of Ca^{2+}.
- The rise in cytoplasmic Ca^{2+} is required for docking of the secretory vesicles in the plasma membrane and for **exocytosis** of the vesicular contents.

(จาก Koeppen and Stanton, 2018. Berne and Levy Physiology. หน้า 691)

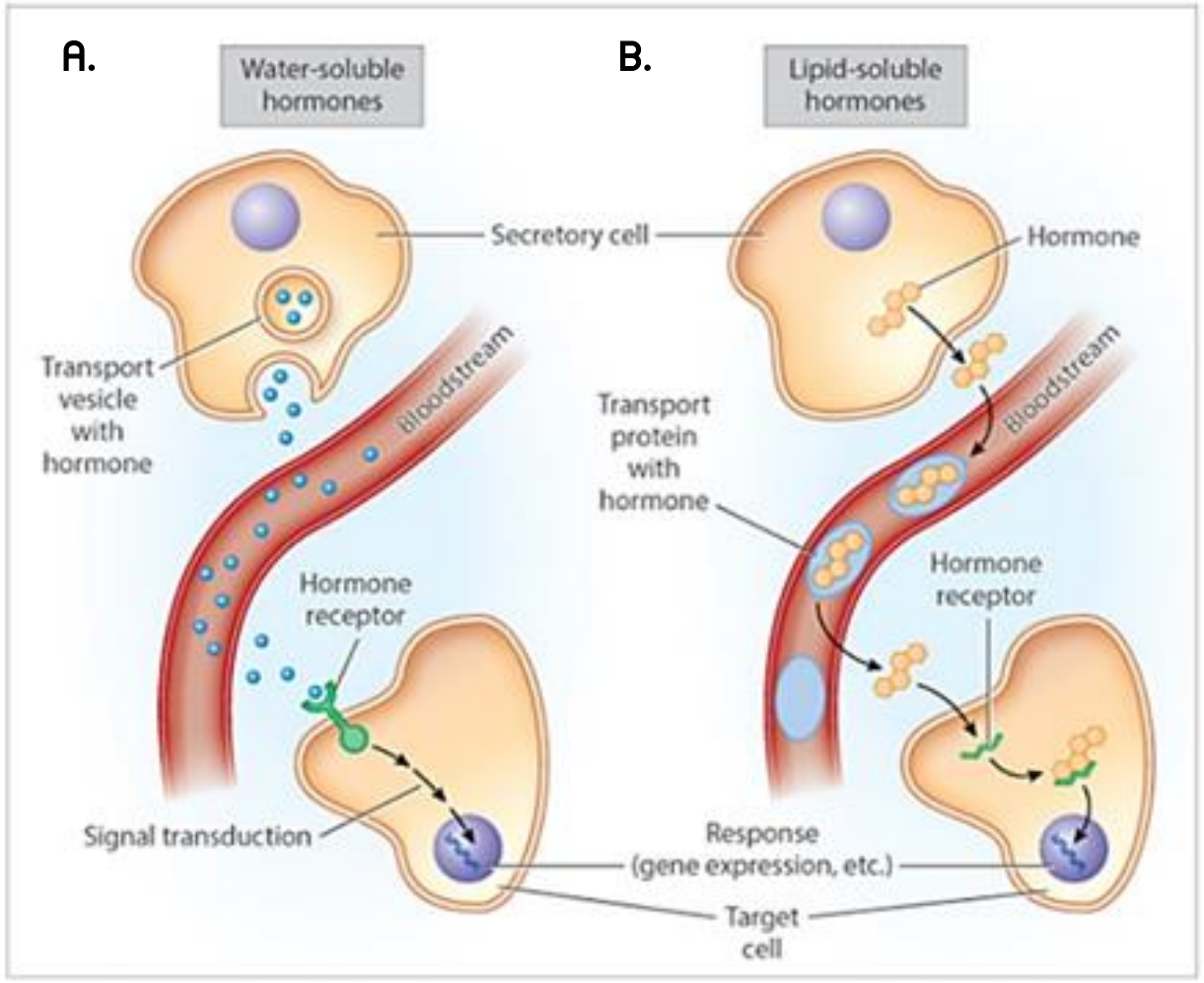
Synthesis and Secretion of Steroid Hormones



- The cell takes up **LDL particle** via the LDL receptor and receptor-mediated endocytosis into **clathrin-coated vesicles**.
- **Lysosomal hydrolases** act on the cholesteryl esters to **release free cholesterol**.
- **Pregnenolone**, the common precursor of all steroid hormones.
- **Pregnenolone** is further metabolized to the major steroid hormones.

(จาก Boron and Boulpaep, 2012. Medical physiology. หน้า 1024)

Transport of hormone in the circulation



A. Majority of amines, peptides, and protein (hydrophilic) hormones circulate in their **free form.**

B. Steroid and thyroid (hydrophobic/lipophilic) hormones circulate bound to specific transport proteins (**binding proteins**).

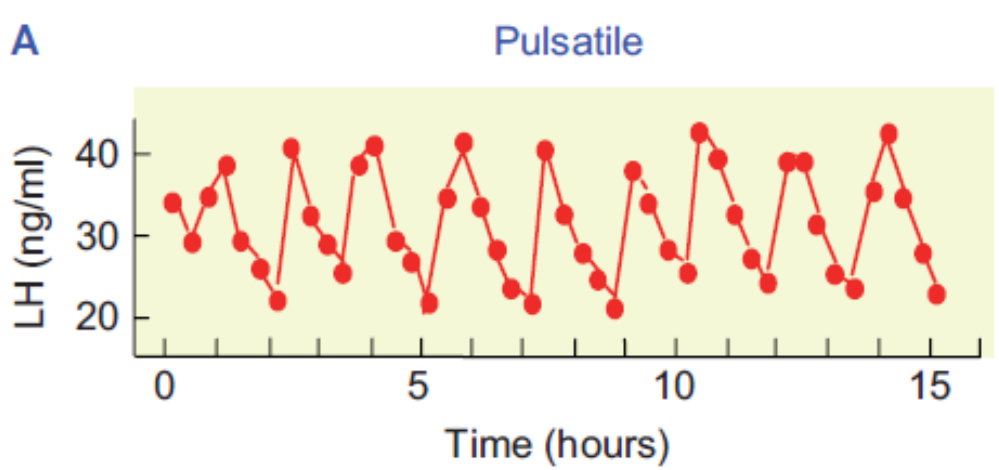
Only unbound hormones (free form) are able to cross the capillary endothelium to reach their sites of biological action or degradation

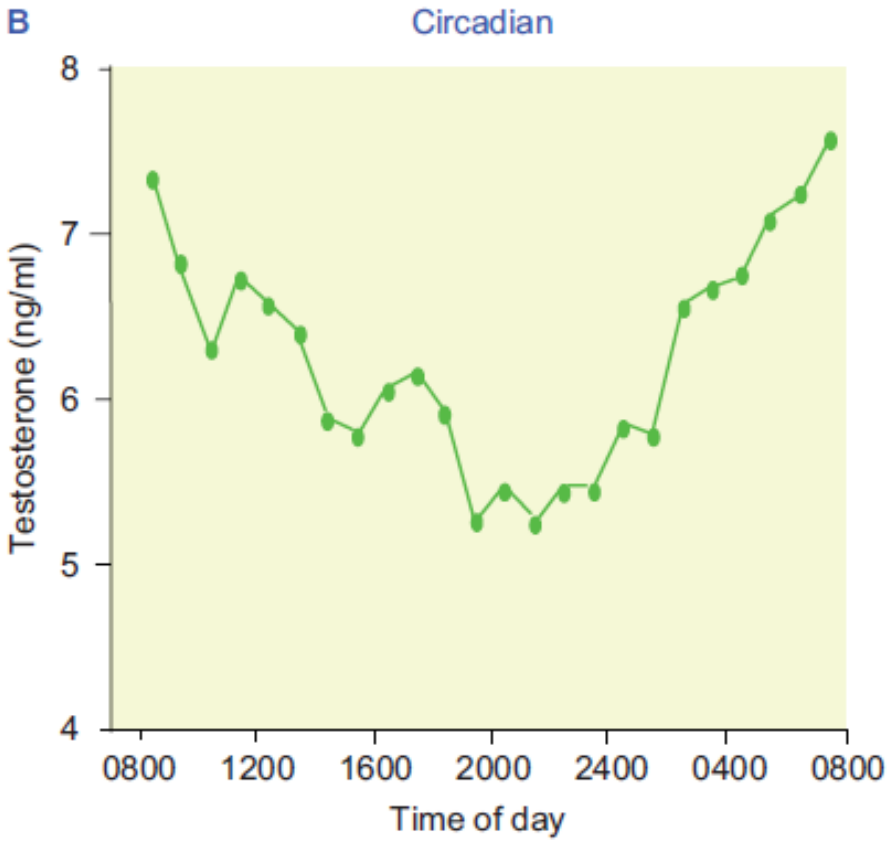
✓ **The binding proteins serve as a reservoir for the hormone and prolong the hormone's half-life**

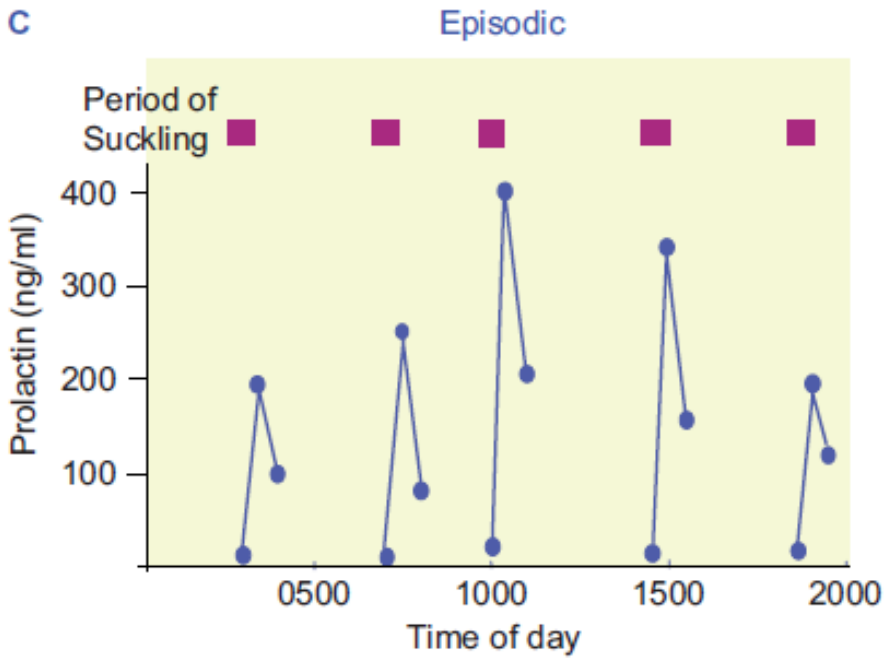
➢ Most carrier proteins are globulins

(https://dls.ym.edu.tw/course/hb/doc/lecture15-(137)%20hormones.pdf)

Hormone levels in blood

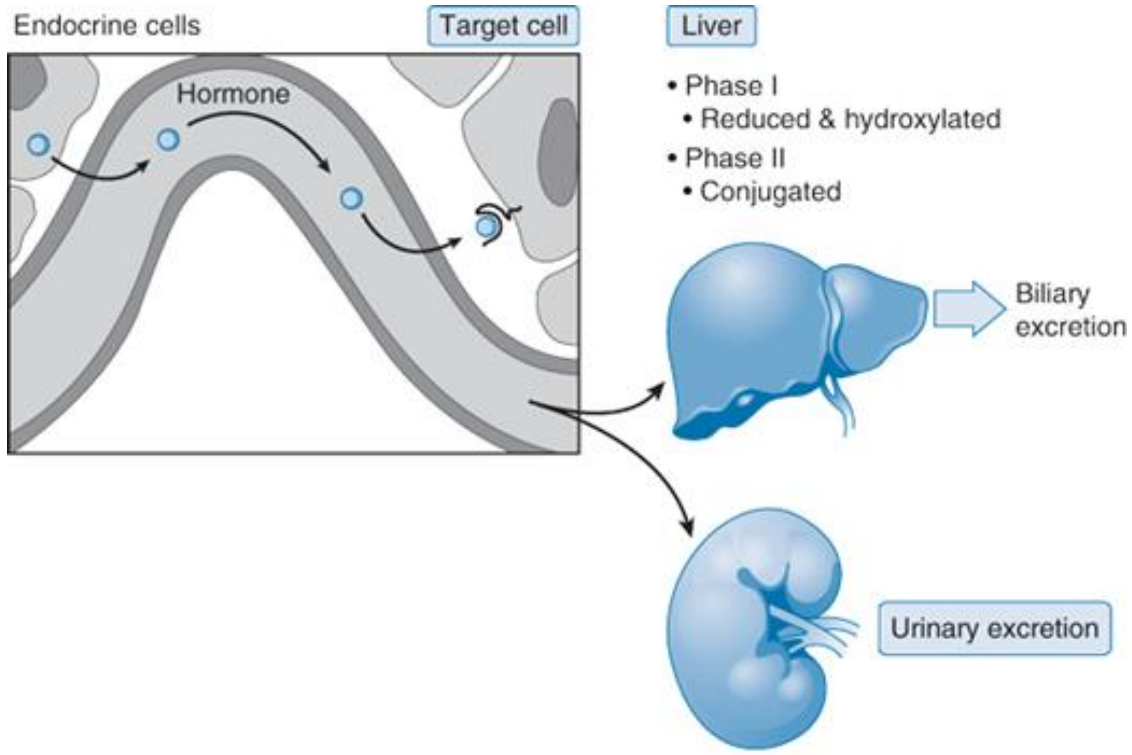






(จาก Goodman, 2009. Basic medical endocrinology. หน้า 26)

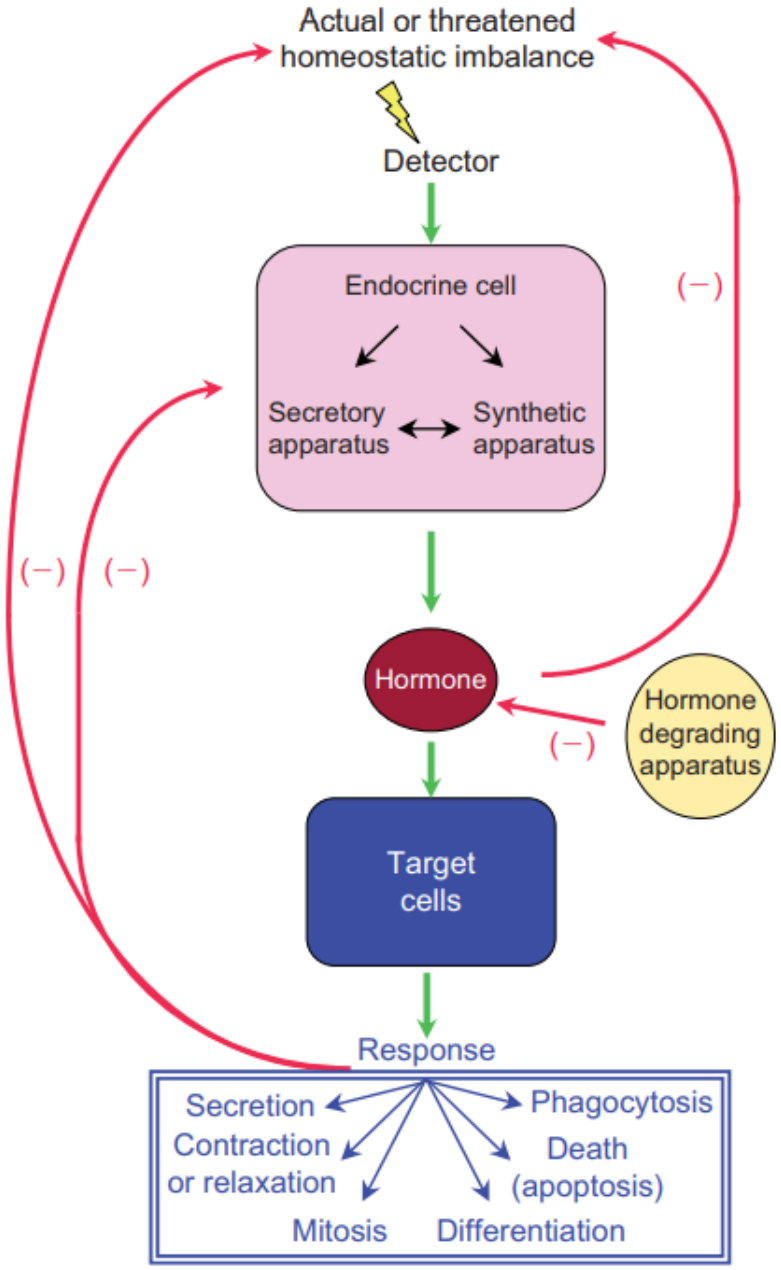
Hormone degradation



Source: Molina PE: *Endocrine Physiology*, 4th Edition: www.accessmedicine.com
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- Only small amount of hormone is degraded after the process of signaling its biological effects
- The remainder are inactivated via **enzymatic degradation**, in blood or intercellular spaces, in liver or kidney cells, as well as the target cells
- Degradation of peptide and protein hormones involves uptake into cells by endocytosis and delivers to cellular sites of degradation, the lysosomes and proteasomes.
- Excretion can be achieved by bile or urinary excretion (via urine and feces)

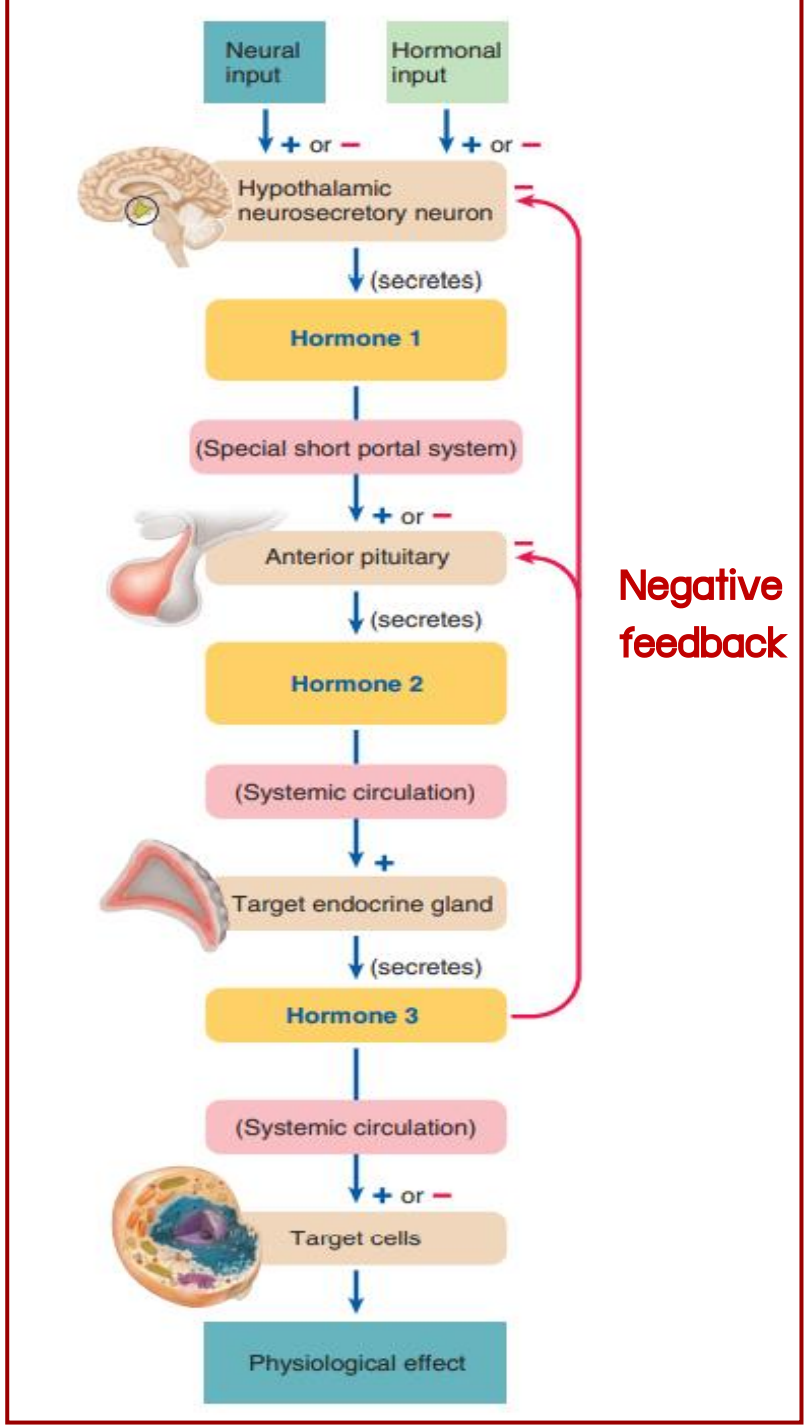
Components of a hormone response system

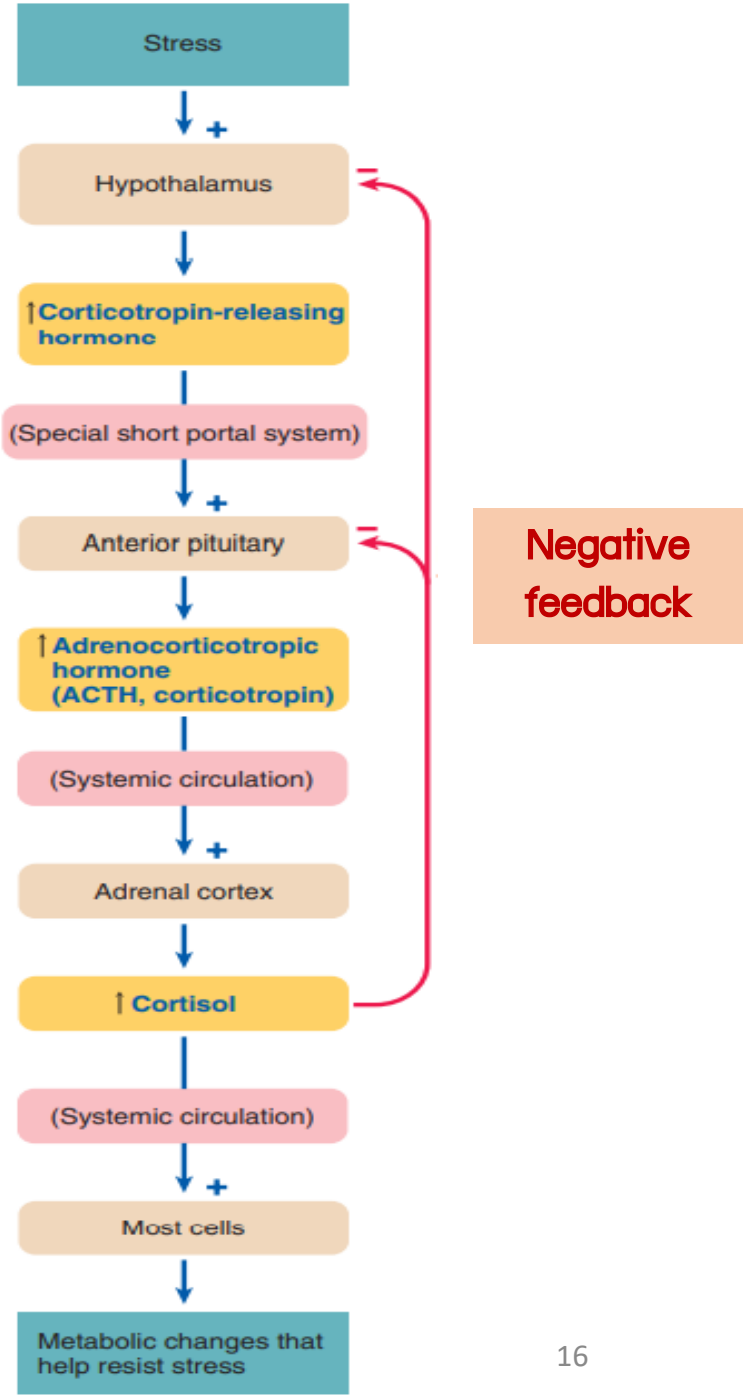


- Negative feedback
- Positive feedback

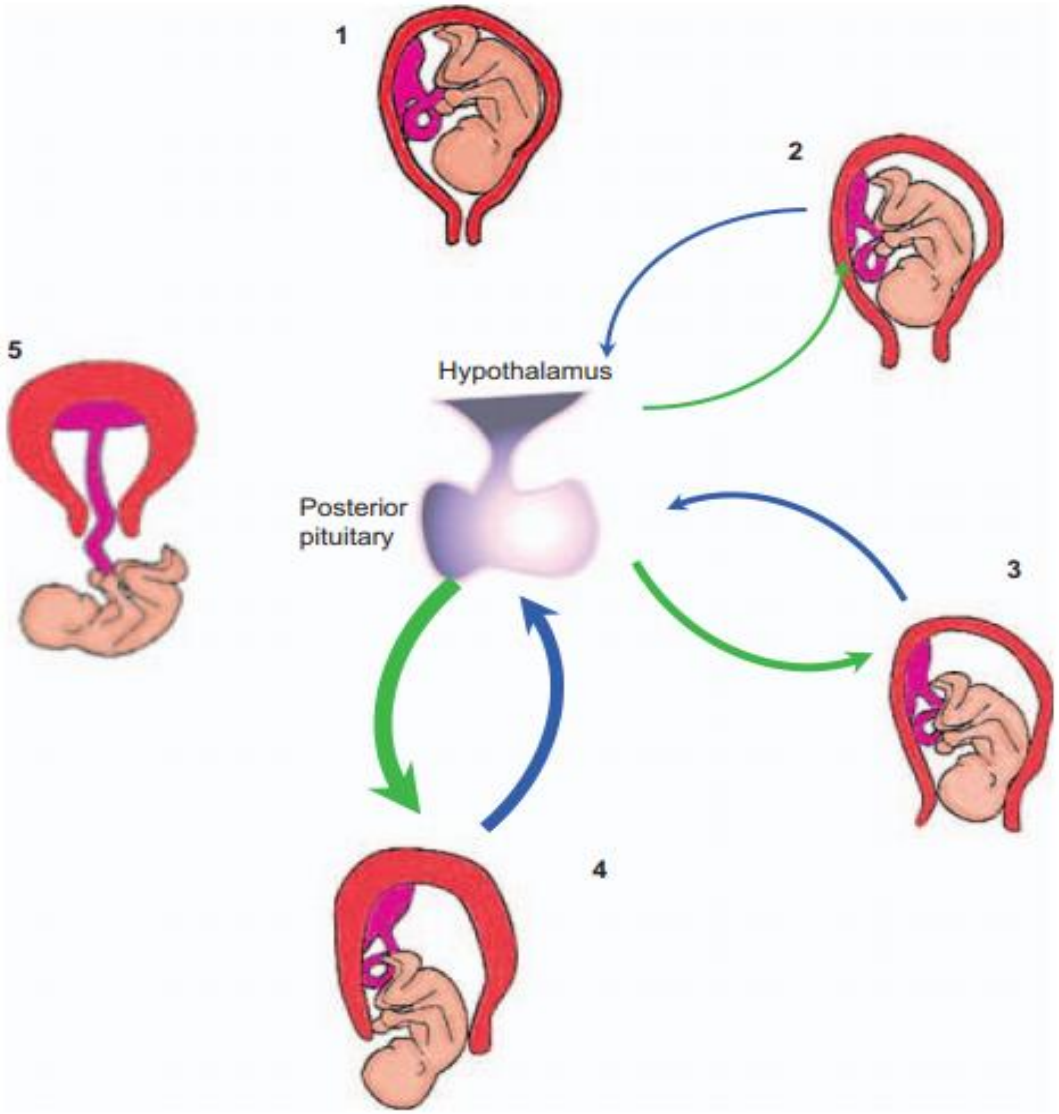
(จาก Goodman, 2009. Basic medical endocrinology. หน้า 22)

Feedback Regulation of Hormone (Negative feedback)





Feedback Regulation of Hormone (Positive feedback)

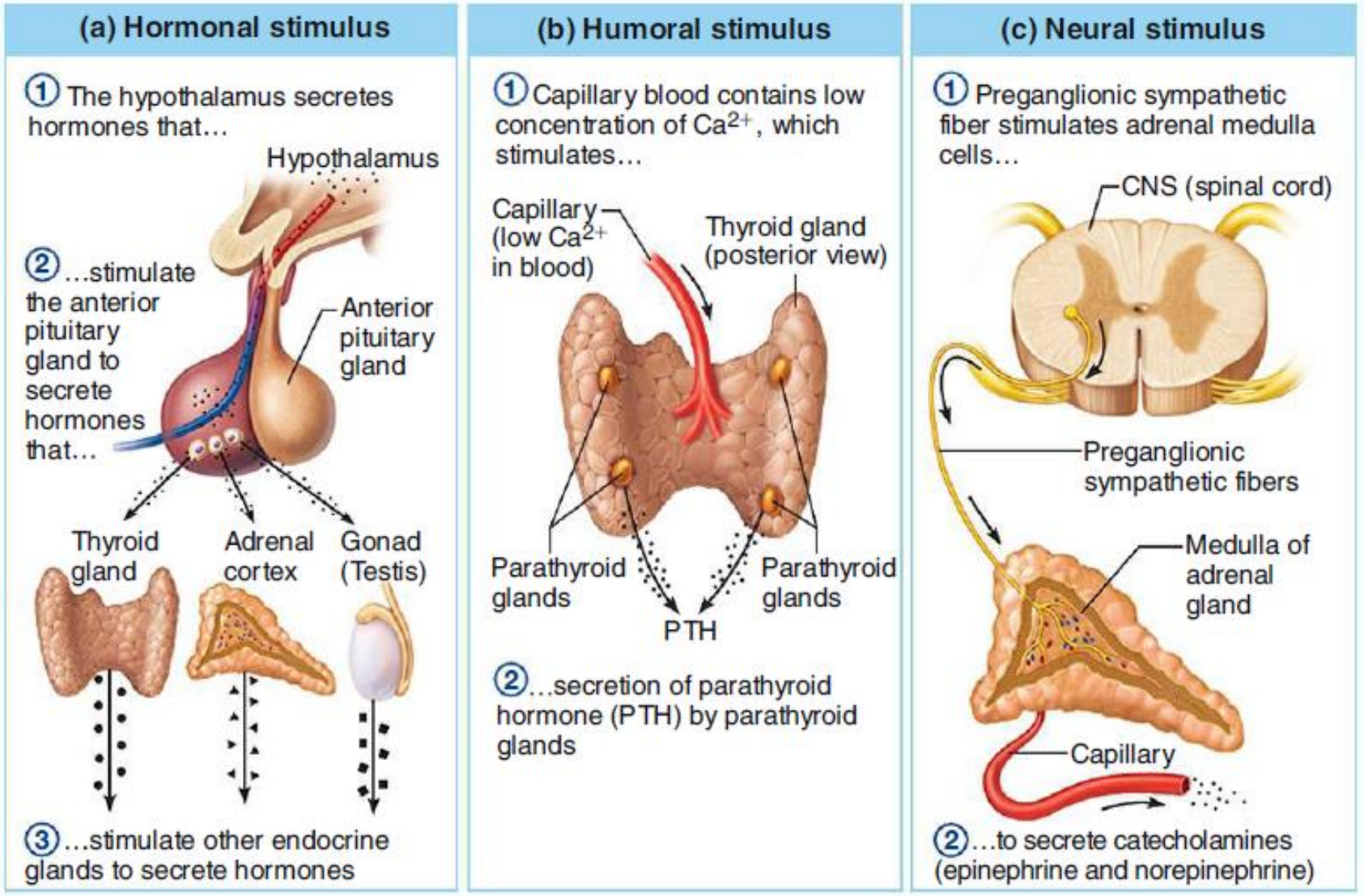


Oxytocin secretion

- Uterine contractions at the onset of parturition apply mild stretch to the cervix
- In response to sensory input from the cervix, oxytocin is secreted from the posterior pituitary gland, and stimulates (+) further contraction of the uterus, which, in turn stimulates secretion of more oxytocin
- Further stretching of the cervix >> more oxytocin secretion until the fetus is expelled

(จาก Goodman, 2009. Basic medical endocrinology. หน้า 24)

Endocrine gland Stimuli



(จาก Marieb, 2012. Essentials human of anatomy and physiology. หน้า 311)

Hormone Receptors

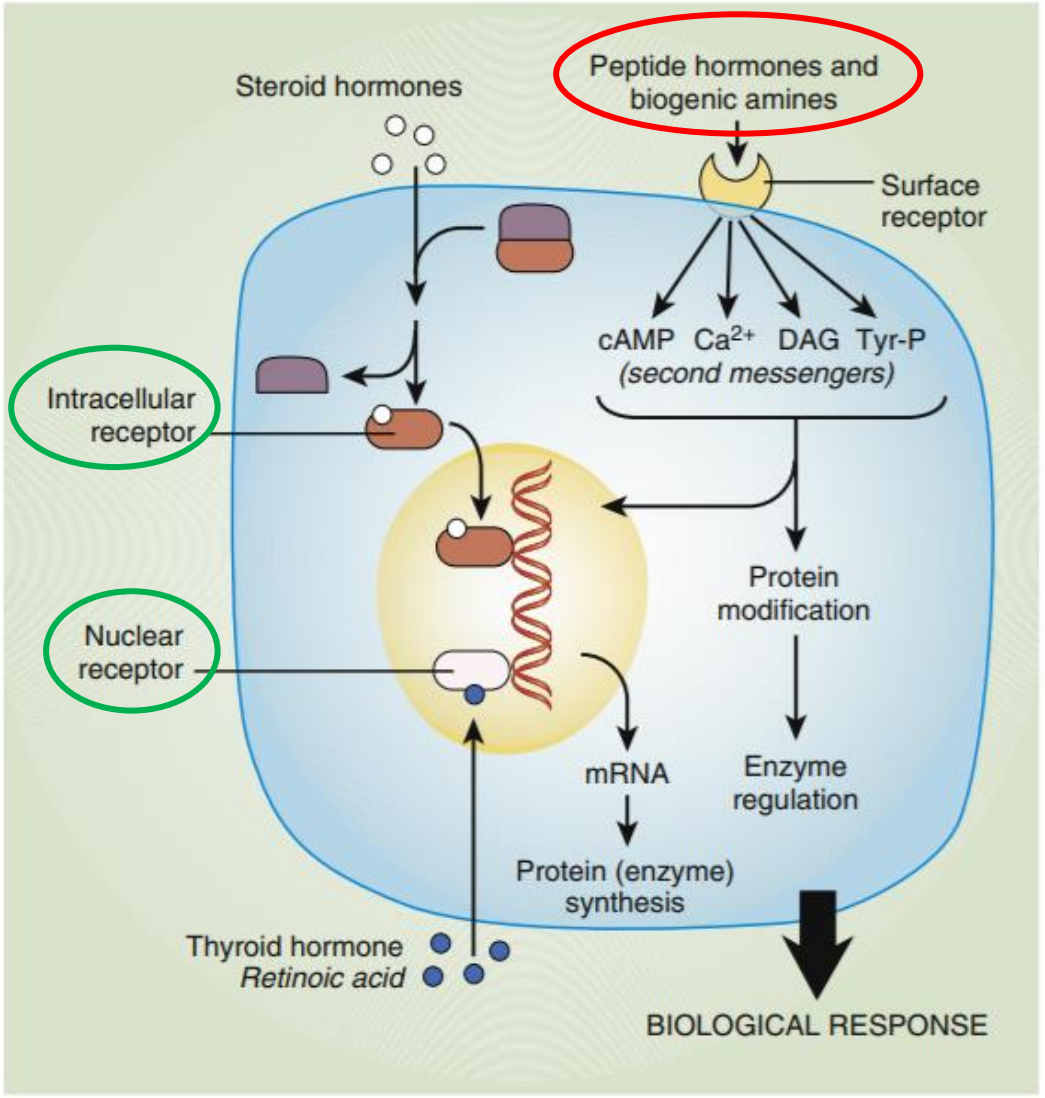


Figure 8.2. Mechanisms by which peptide and steroid hormones signal. Ca^{2+}, calcium; cAMP, cyclic adenosine monophosphate; DAG, diacylglycerol; mRNA, messenger RNA; Tyr-P, phosphorylated tyrosine residue. *(From Goldman L, Ausiello D.* Cecil Textbook of Medicine. *22nd ed. Philadelphia: WB Saunders; 2004.)*

Hormone Receptors: Cell-surface receptors

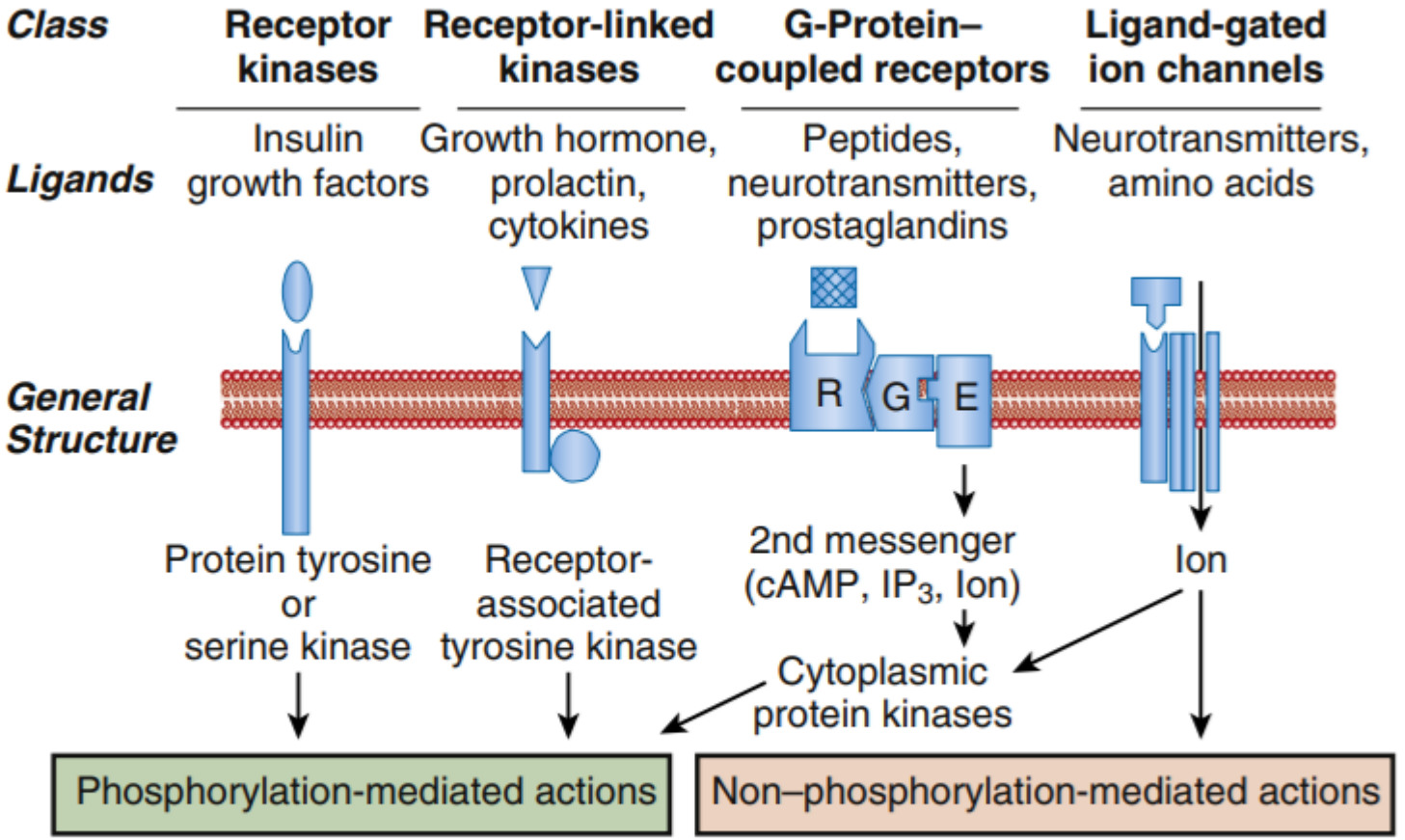
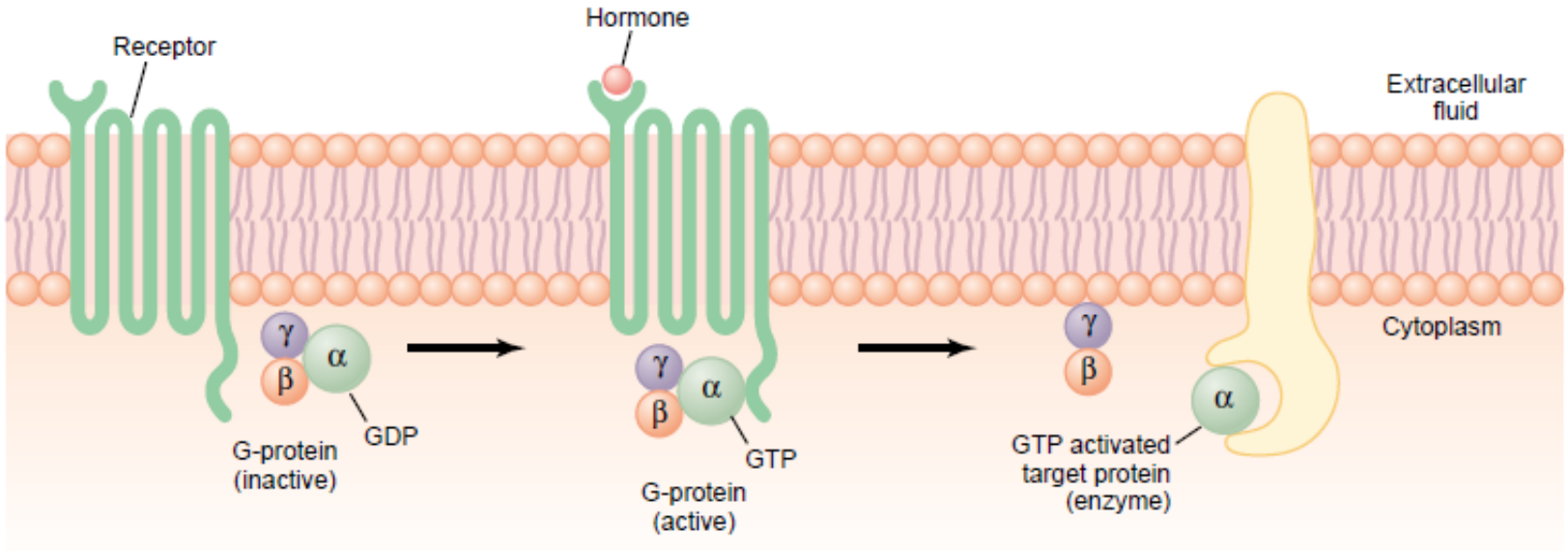
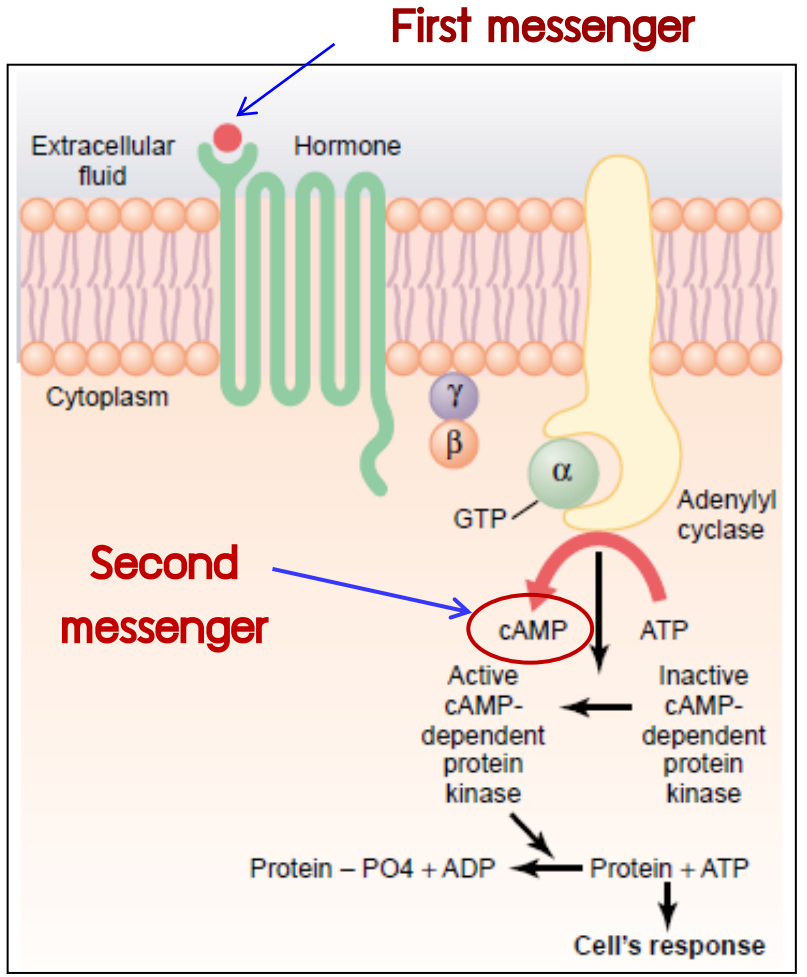


Figure 8.3. The four major classes of membrane receptors for hormones and neurotransmitters. cAMP, cyclic adenosine monophosphate; E, effector enzyme; G, G protein; IP_3, inositol triphosphate; R, receptor. *(From Brown TA, Brown D.* USMLE Step 1 Secrets. *Philadelphia: Hanley & Belfus; 2004.)*

Mechanism of G protein-coupled receptor activation





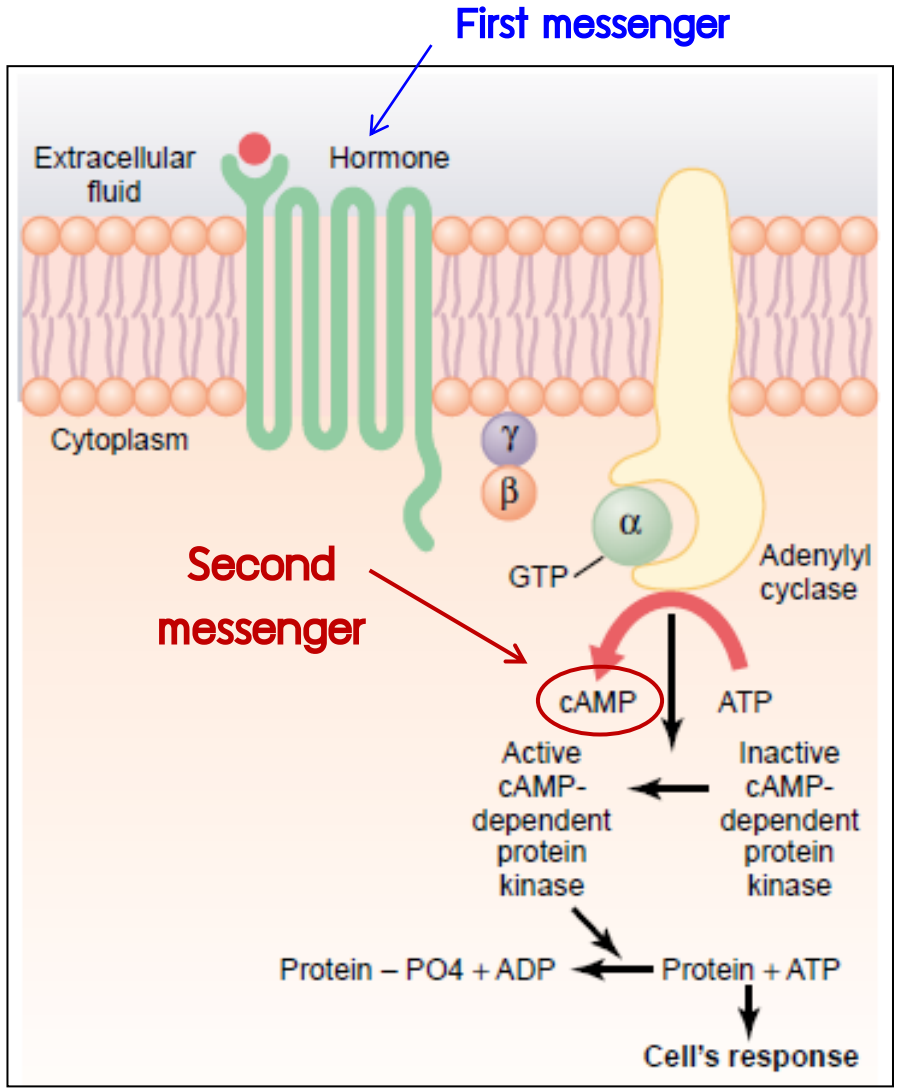
- When the hormone activates the receptor, the inactive α, β, and γ G protein complex associates with the receptor and is activated, with an exchange of guanosine triphosphate (GTP) for GDP.

- This causes the α subunit (to which the GTP is bound) to dissociate from the β and γ subunits of the G protein and to interact with membrane-bound target enzymes **(Adenylyl cyclase)** that initiate intracellular signals.

(จาก Hall, 2011. Guyton and Hall Textbook of Medical Physiology. หน้า 911)

Second messengers

- Short lived intracellular signaling molecules
- Elevated second messenger concentration leads to rapid alteration in the activity of cellular enzymes
- Removal or degradation of second messenger terminate the cellular response
- Example of second messengers
  - Cyclic nucleotides (cAMP, cGMP)
  - Ca^{2+}
  - Inositol trisphosphate (IP_3)



(จาก Hall, 2011. Guyton and Hall Textbook of Medical Physiology. หน้า 913)

Receptors and downstream effectors for Peptide hormones

Agonists	Receptor	Linked Enzyme	Second Messenger
PTH	Coupled to $G\alpha_s$	Adenylyl cyclase	cAMP
ANG II	Coupled to $G\alpha_i$	Adenylyl cyclase (inhibited)	cAMP
AVP, ANG II, TRH	Coupled to $G\alpha_q$	PLC	IP_3 and DAG
ANG II	Coupled to G_i/G_o	PLA_2	Arachidonic acid metabolites
ANP	Guanylyl cyclase	Guanylyl cyclase	cGMP
Insulin, IGF-1, IGF-2, EGF, PDGF	Tyrosine kinase	Tyrosine kinase	Phosphoproteins
GH, erythropoietin, LIF	Associated with tyrosine kinase	JAK-STAT family of tyrosine kinases	Phosphoproteins

ANG II, angiotensin II; ANP, atrial natriuretic peptide; EGF, epidermal growth factor; JAK-STAT, Janus kinase/signal transducer and activator of transcription; LIF, leukemia inhibitory factor.

IP_3 , Inositol trisphosphate

DAG, Diacylglycerol

(จาก Boron and Boulpaep, 2012. Medical physiology. หน้า 1021)

Receptors and downstream effectors for Peptide hormones

Peptide hormones bind to cell surface receptors and activate a variety of signal transduction systems

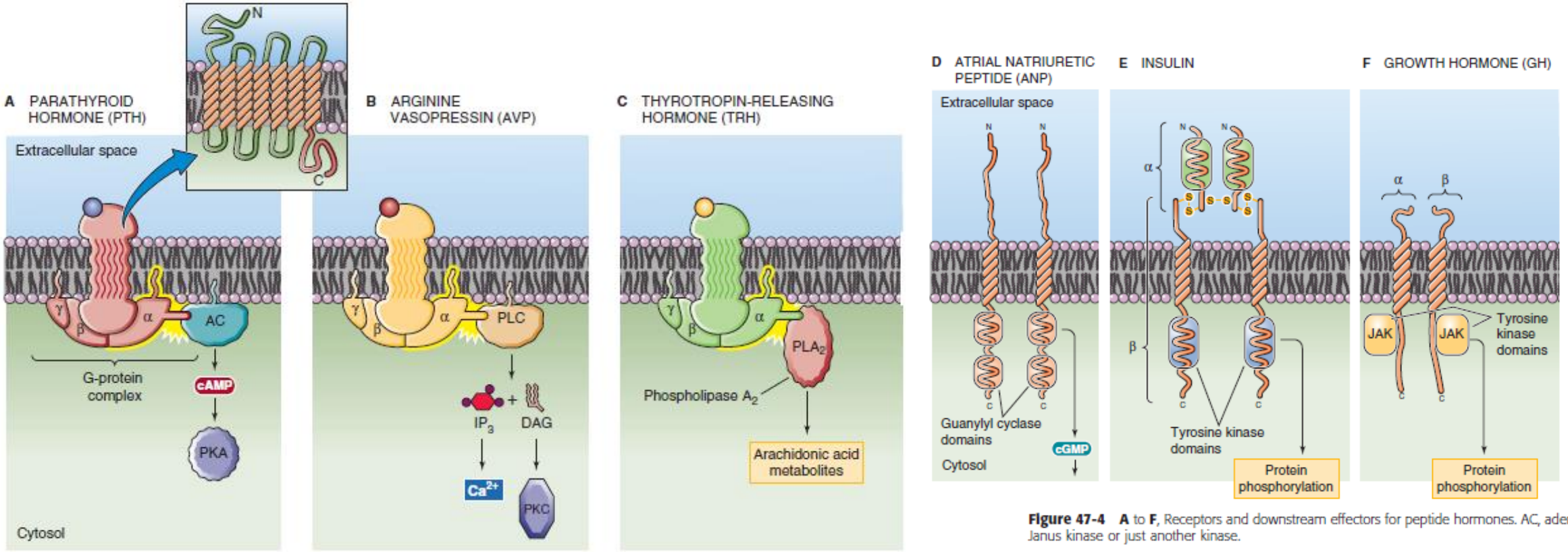
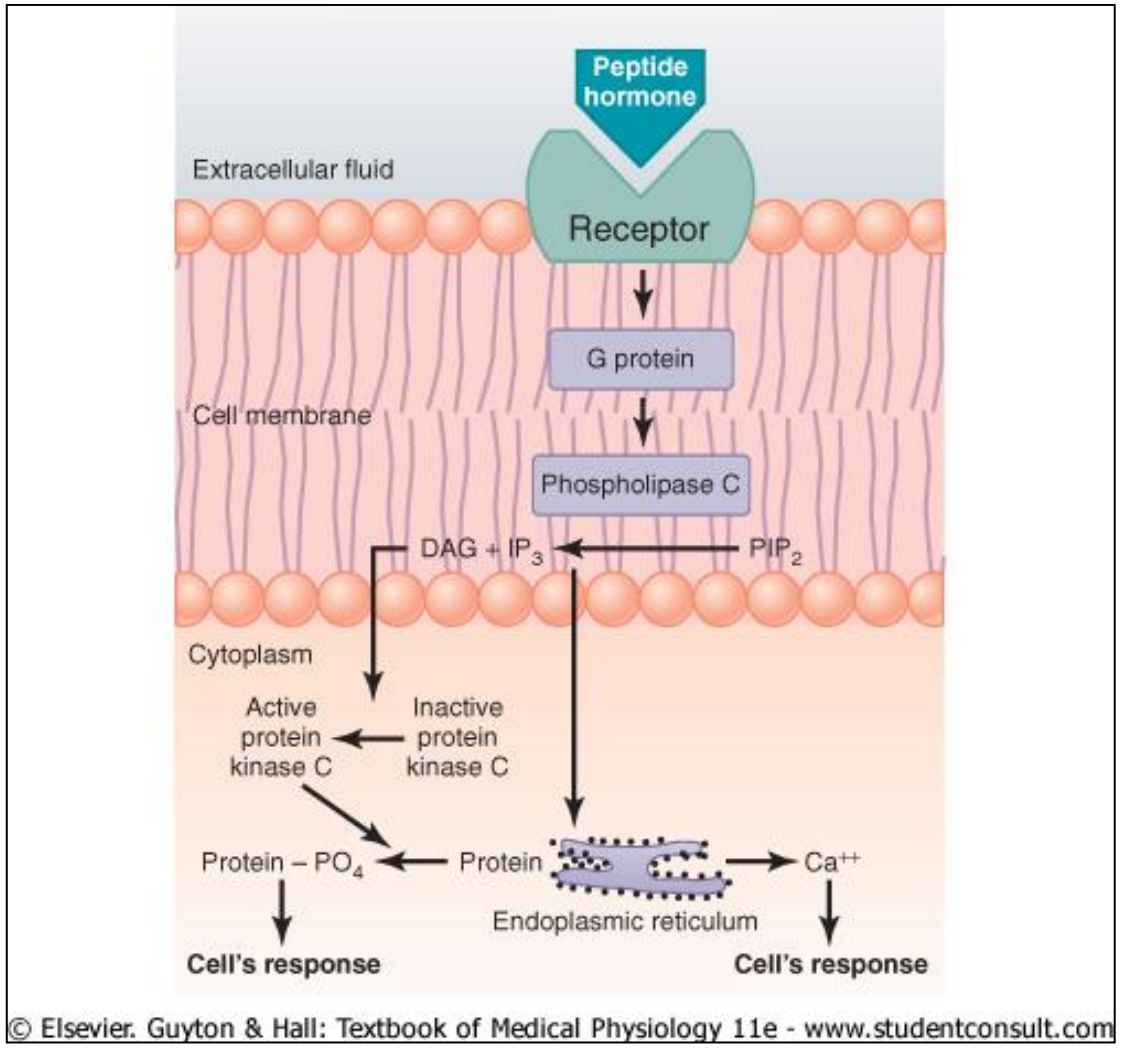


Figure 47-4 **A** to **F**, Receptors and downstream effectors for peptide hormones. AC, adenylyl cyclase; JAK, Janus kinase or just another kinase.

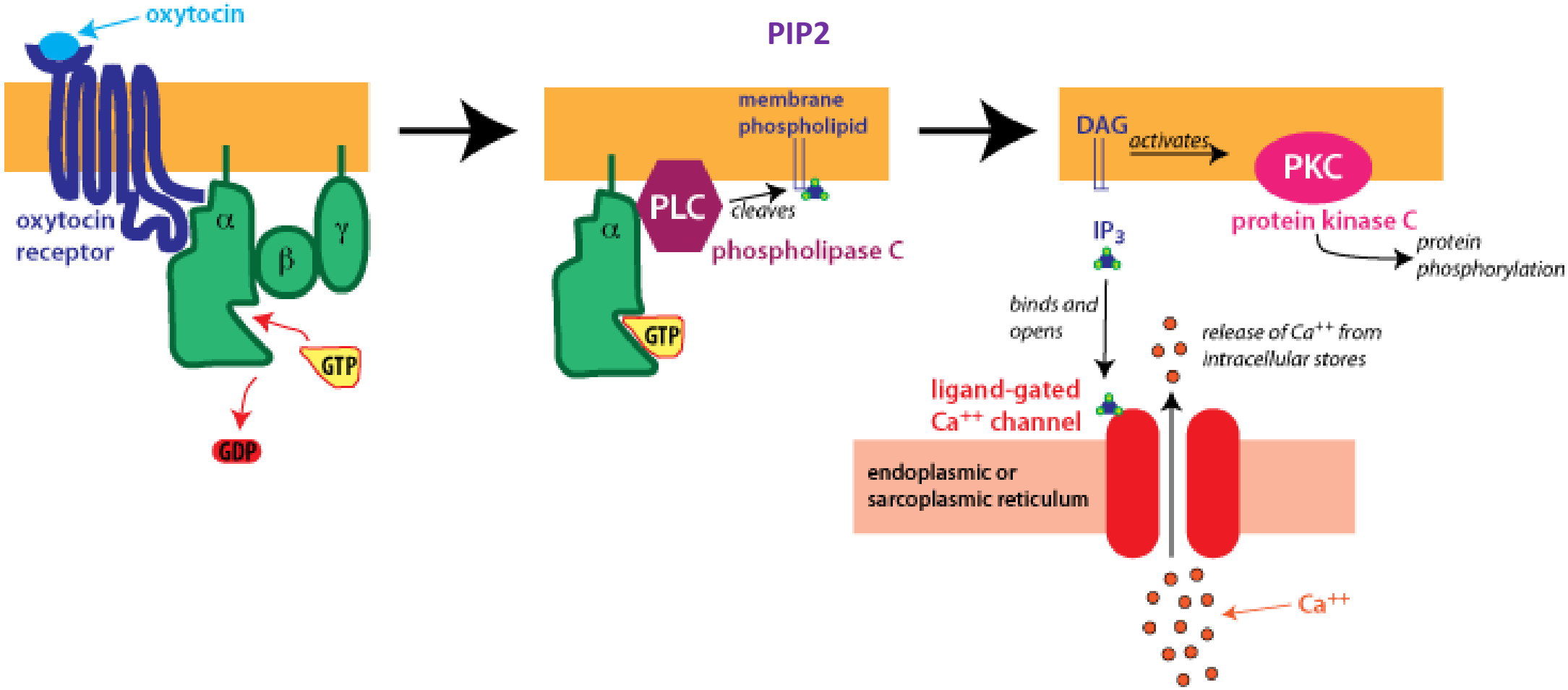
(Boron and Boulpaep, 2012)

G Proteins Coupled to Phospholipase C



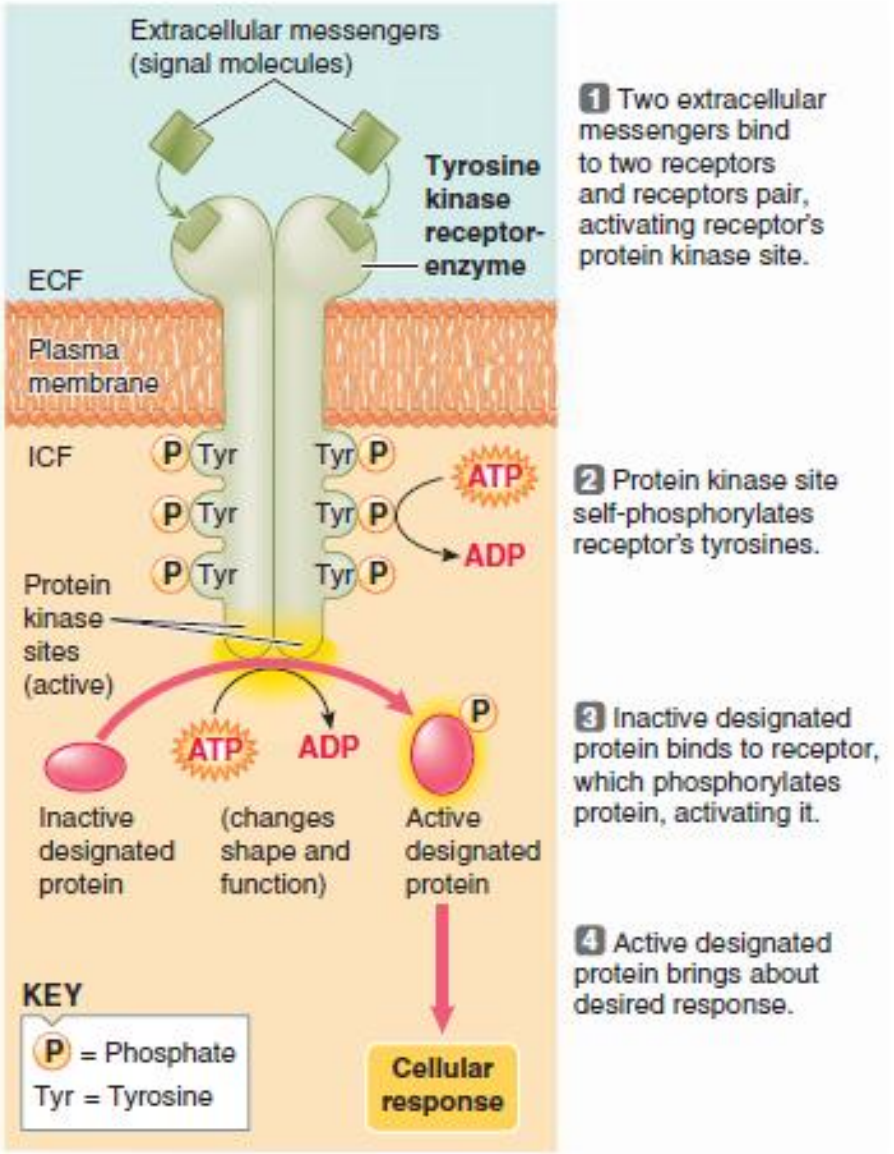
The cell membrane phospholipid second messenger system by which some hormones exert their control of cell function. DAG, diacylglycerol; IP3, inositol triphosphate; PIP2, phosphatidylinositol biphosphate.

G Proteins Coupled to Phospholipase C
Oxytocin receptor on smooth muscle



(https://www.youtube.com/watch?v=wC2_7Ror3qY)

Mechanism of tyrosine kinase receptors

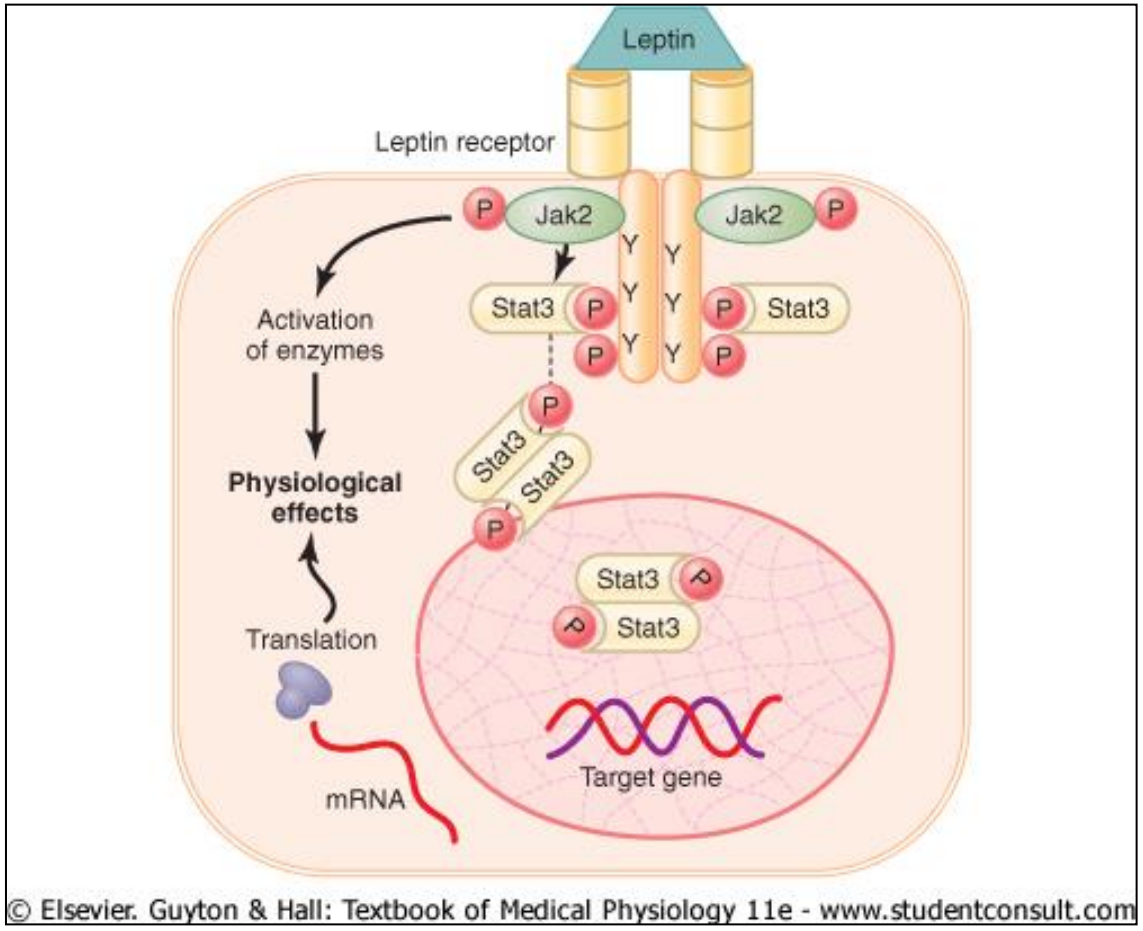


Binding of hormones
>> Receptor dimerization

(Autophosphorylation)

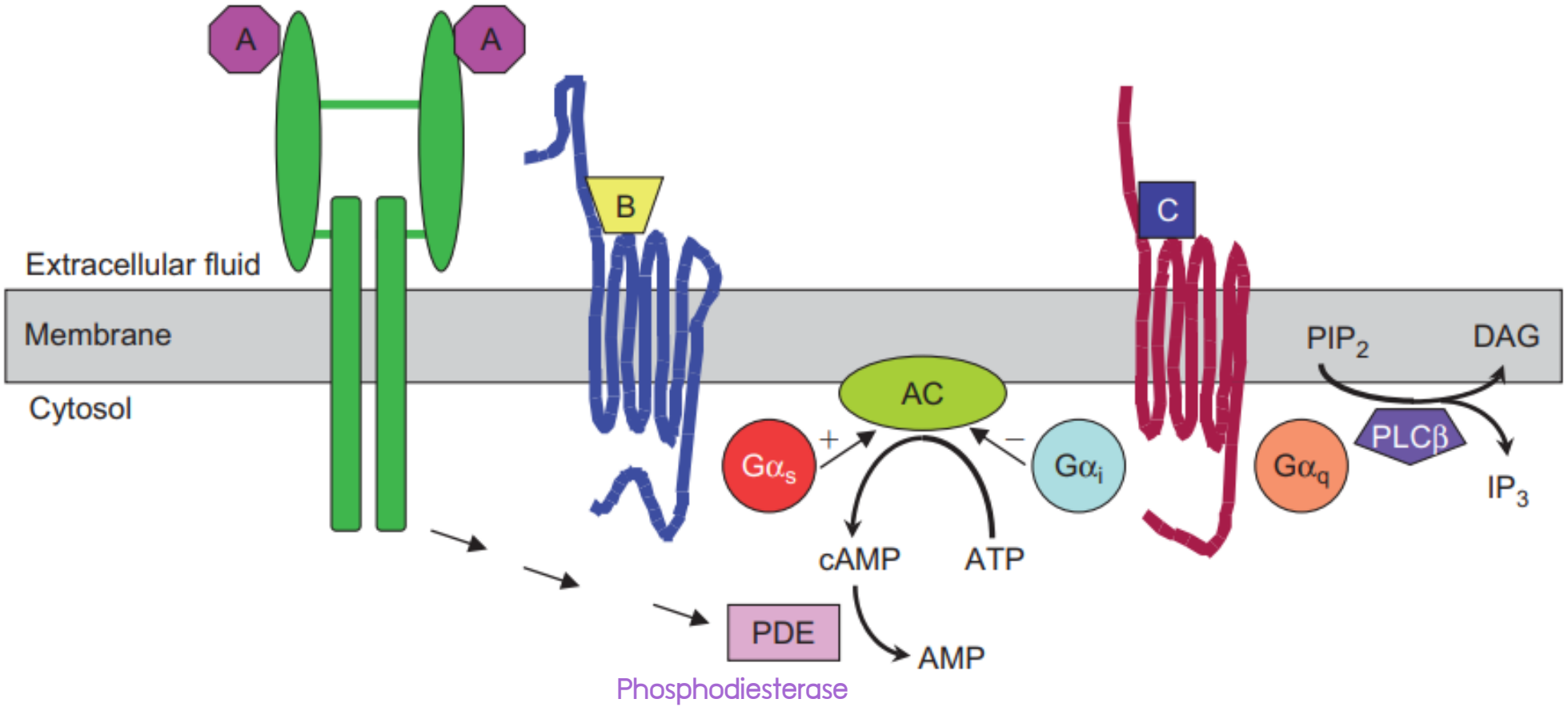
(จาก Sherwood, 2016. Human physiology. หน้า 117)

Tyrosine Kinase-Associated Receptors



An enzyme-linked receptor-the leptin receptor. The receptor exists as a homodimer (two identical parts), and leptin binds to the extracellular part of the receptor, causing phosphorylation and activation of the intracellular associated janus kinase 2 (JAK2). This causes phosphorylation of signal transducer and activator of transcription (STAT) proteins, which then activates the transcription of target genes and the synthesis of proteins. JAK2 phosphorylation also activates several other enzyme systems that mediate some of the more rapid effects of leptin.

Integration of Hormonal Actions at the Cellular Level



Cells may receive inputs from hormones A, B, and C simultaneously. Hormone B acting through a G-protein coupled receptor activates adenylyl cyclase through the α stimulatory subunit G αs . Hormone C binds to its G-protein coupled receptor, which inhibits adenylyl cyclase through the inhibitory subunit G αi , and activates phospholipase C through the G αq subunit, resulting in cleavage of phosphatidylinositol bis-phosphate (PIP2) and the release of diacylglycerol (DAG) and inositol trisphosphate (IP3). Hormone A, acting through a tyrosine kinase receptor, activates cyclic AMP phosphodiesterase (PDE), which degrades cAMP. The combined actions of the three hormones determines the concentration of cyclic AMP

(จาก Goodman, 2009. Basic medical endocrinology. หน้า 92)

Action of Steroid hormones

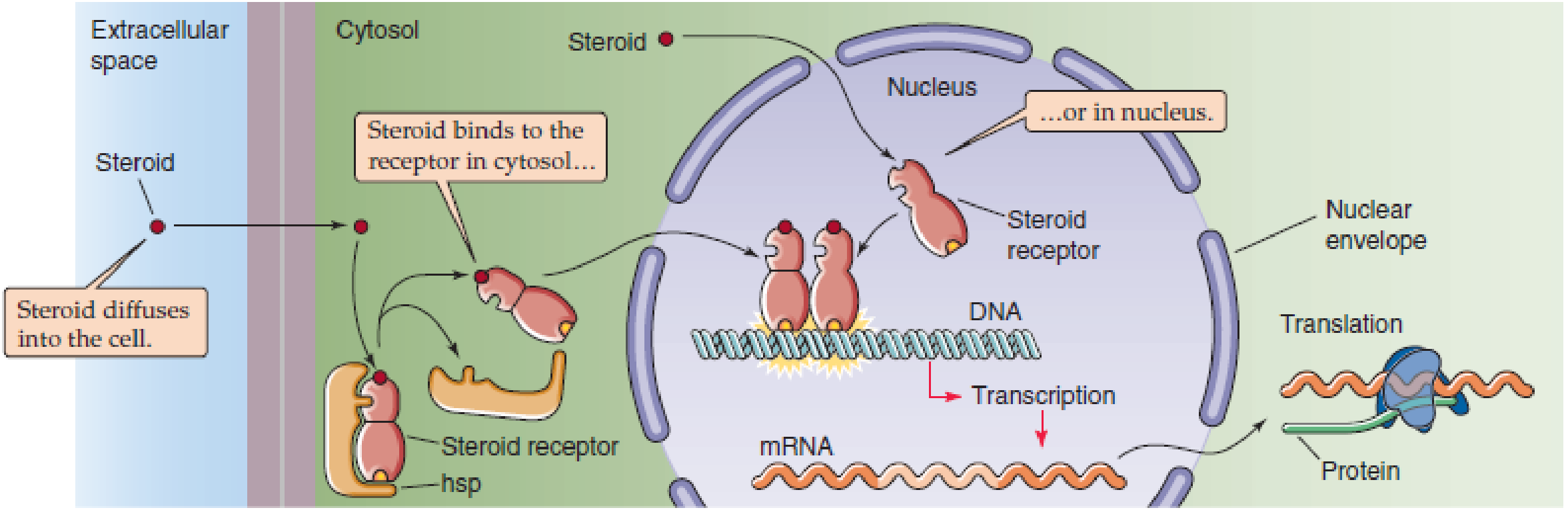


Figure 47-7 Action of steroid hormones. The activated steroid hormone receptor binds to specific stretches of DNA called SREs, thus stimulating the transcription of appropriate genes. hsp, heat shock protein.

(Boron and Boulpaep, 2012)

Table 47-4 Differences Between Steroid and Peptide Amine Hormones

Property	Steroid Hormones	Peptide Amine Hormones
Storage pools	None	Secretory vesicles
Interaction with cell membrane	Diffusion through cell membrane	Binding to receptor on cell membrane
Receptor	In cytoplasm or nucleus	On cell membrane
Action	Regulation of gene transcription (primarily)	Signal transduction cascades affecting a variety of cell processes
Response time	Hours to days (primarily)	Seconds to minutes

(Boron and Boulpaep, 2012)

Hypothalamus and Pituitary

Hypothalamus

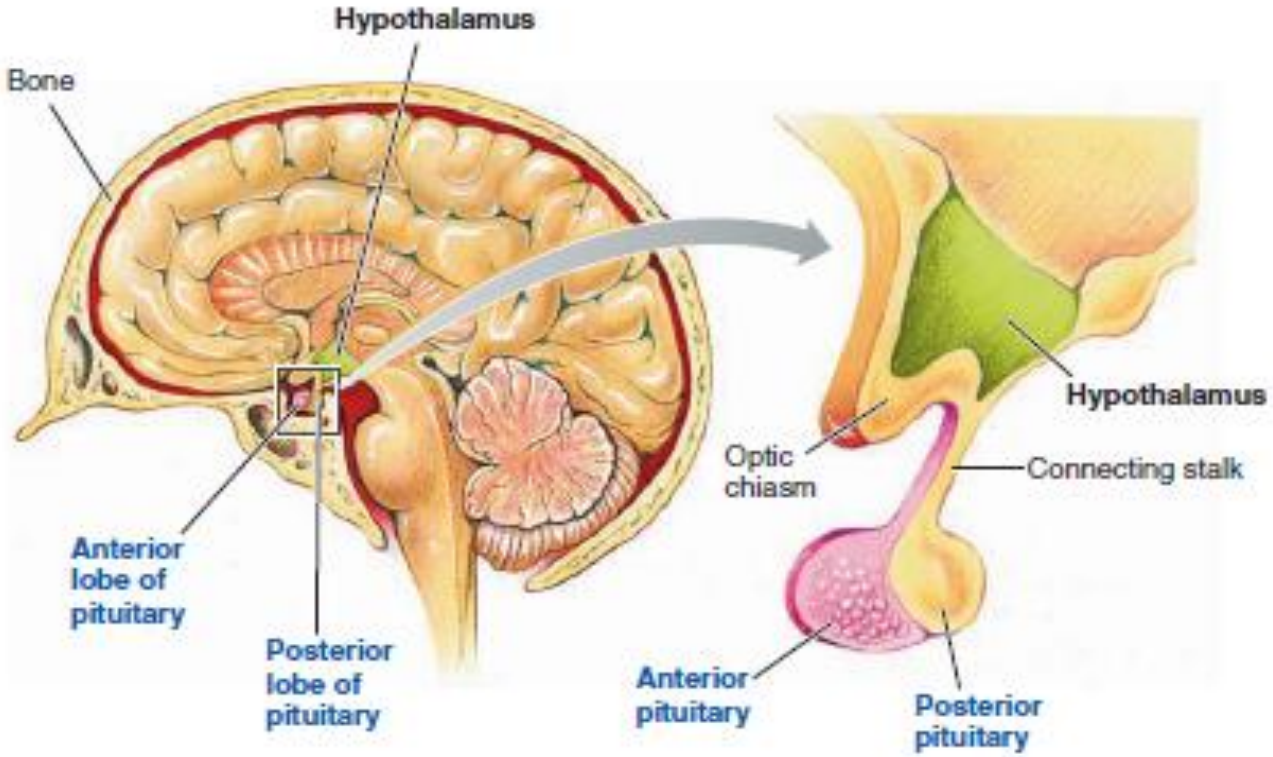
- coordinating the physiologic responses to maintain homeostasis; food intake, energy expenditure, body weight, sleep cycle

Pituitary gland (hypophysis)

- located in a bony cavity at the base of the brain below the hypothalamus
- connected to the hypothalamus by connecting stalk.

Posterior pituitary : nervous tissue >> **neurohypophysis.**

Anterior pituitary : glandular epithelial tissue
>> **adenohypophysis** (*adeno* means "glandular").



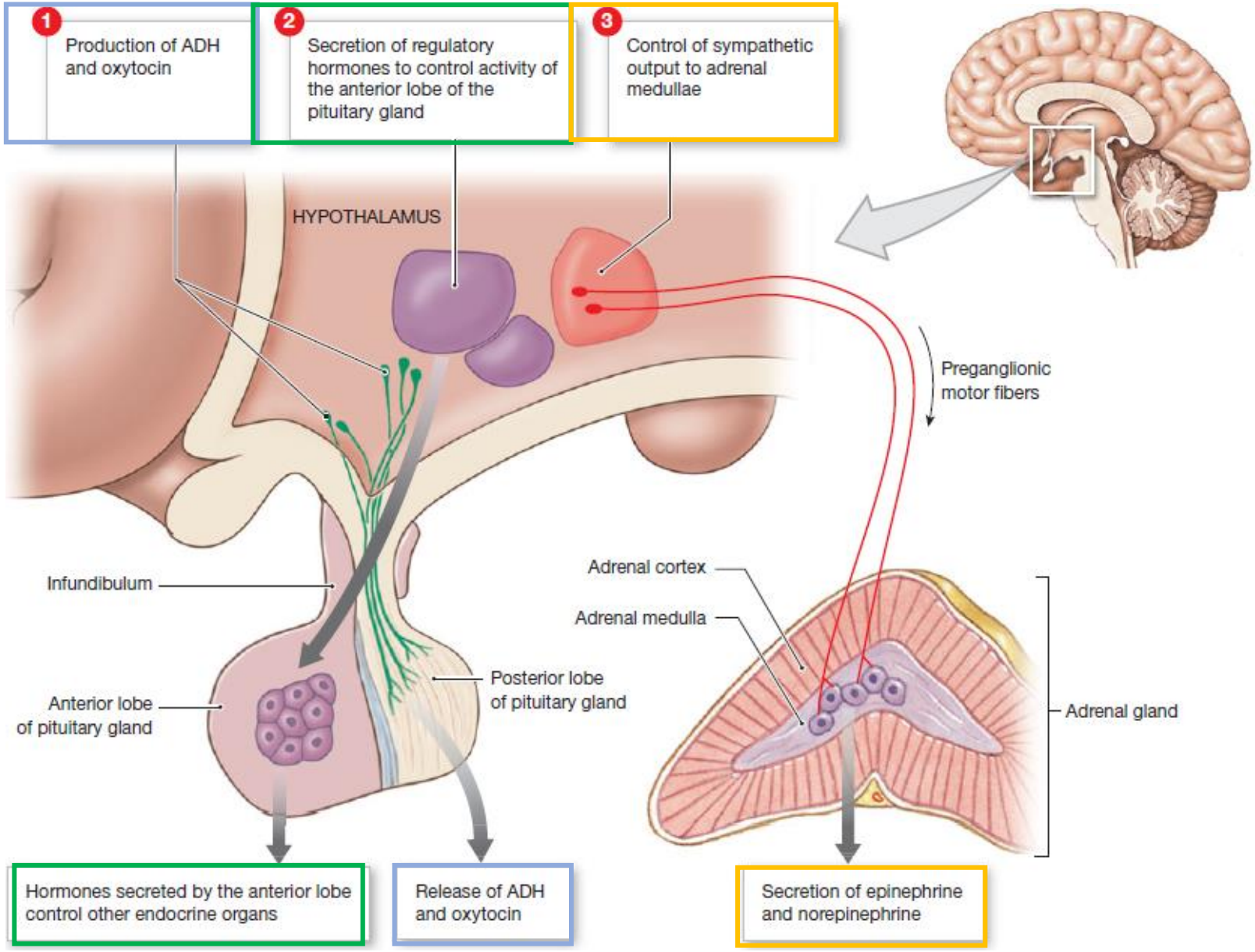
(a) Relation of pituitary gland to hypothalamus and rest of brain

(b) Enlargement of pituitary gland and its connection to hypothalamus

● **FIGURE 18-4 Anatomy of the pituitary gland.**

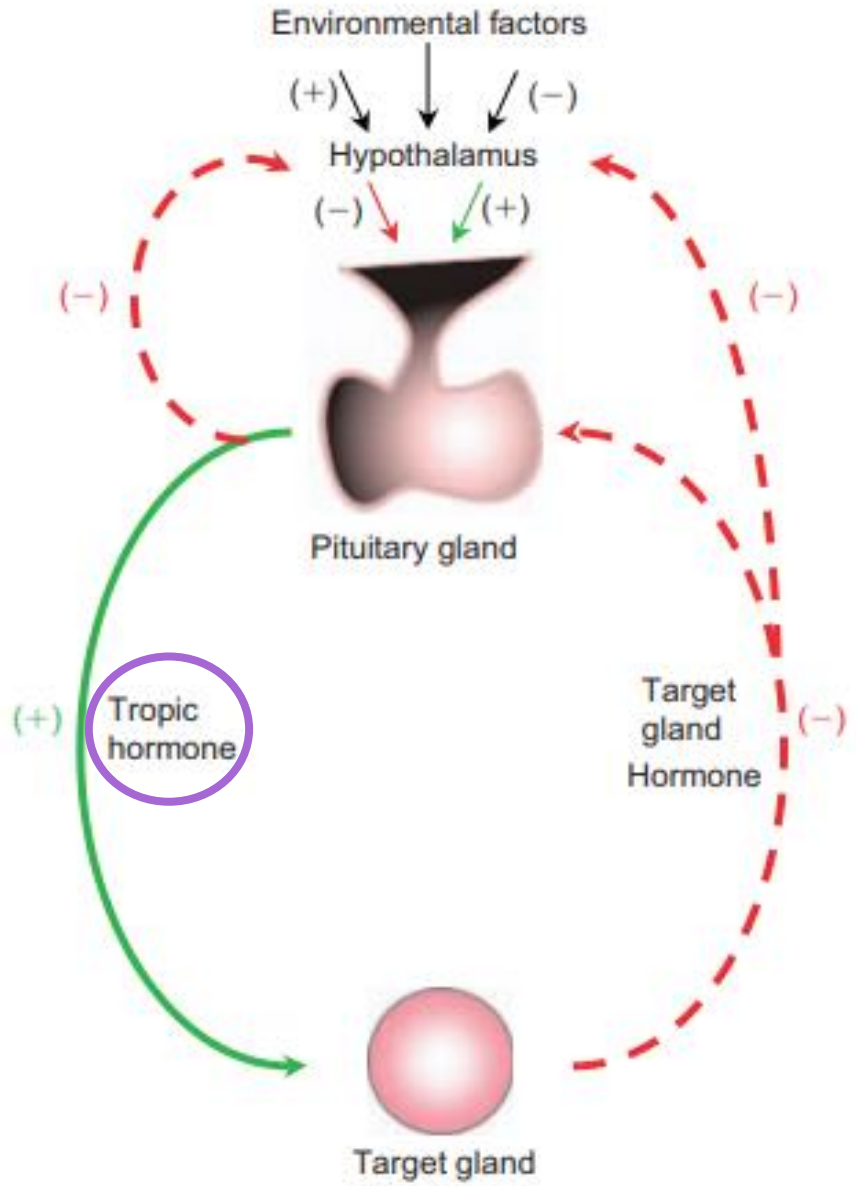
(จาก Sherwood, 2016. Human physiology. หน้า 646)

Mechanisms of hypothalamic control endocrine function



(Martini et al., 2012)

Hypothalamic-pituitary-axis



- Environmental factors may increase or decrease pituitary activity by increasing or decreasing hypophysiotropic hormone secretion.
- Pituitary secretions increase the secretion of target gland hormones, which may inhibit further secretion by acting at either the hypothalamus or the pituitary.

Tropic >> "nourishing"

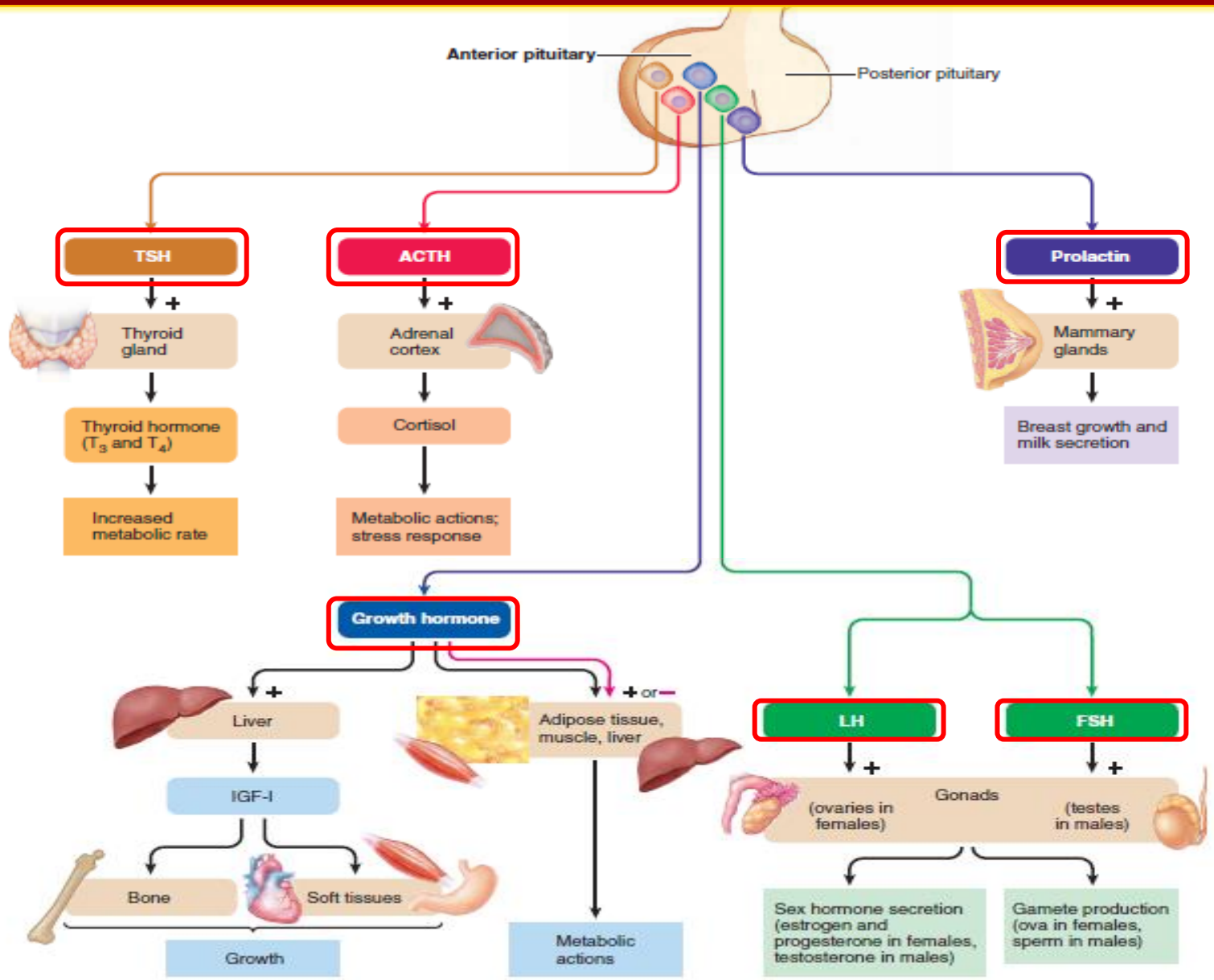
A hormone that has as its primary function the regulation of hormone secretion by another endocrine gland.

Example

thyroid-stimulating hormone (TSH) from the anterior pituitary, stimulates thyroid hormone secretion by the thyroid gland and also maintains the structural integrity of this gland.

(จาก Goodman, 2009. Basic medical endocrinology. หน้า 39)

Functions of anterior pituitary hormones



(จาก Sherwood, 2016. Human physiology. หน้า 649)

Functions of the anterior pituitary hormones

ANTERIOR PITUITARY			
Releasing (Inhibitory) Factor Made by Hypothalamus	**Target Cell in Anterior Pituitary**	**Hormone Released by Anterior Pituitary**	**Target of Anterior Pituitary Hormone**
GHRH (inhibited by somatostatin)	Somatotroph	GH	Stimulates IGF-1 production by multiple somatic tissues, especially liver
TRH	Thyrotroph	TSH	Thyroid follicular cells, stimulated to make thyroid hormone
CRH	Corticotroph	ACTH	Fasciculata and reticularis cells of the adrenal cortex, to make corticosteroids
GnRH	Gonadotroph	FSH	Ovarian follicular cells, to make estrogens and progestins Sertoli cells, to initiate spermatogenesis
GnRH	Gonadotroph	LH	Leydig cells, to make testosterone
(inhibited by dopamine)	Lactotroph	PRL	Mammary glands, to initiate and maintain milk production

(จาก Boron and Boulpaep, 2012. Medical physiology. หน้า 1017)

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